

CIVL6410 Transport Network Modelling

Week 5: Traffic Assignment Problems

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Semester 1, 2026

Notes for This Week's Python Task

- ▶ This week's tutorial (Python task) uses ***Conditional Filtering*** in Pandas DataFrames.
- ▶ The guidance document *Blackboard > Content > Python Programming > 'pandas_numpy_basics.pdf'* has been updated to include a new section, '***8. Selecting Rows with Boolean Filtering***' for your review.

Outline

► Traffic Assignment Problems

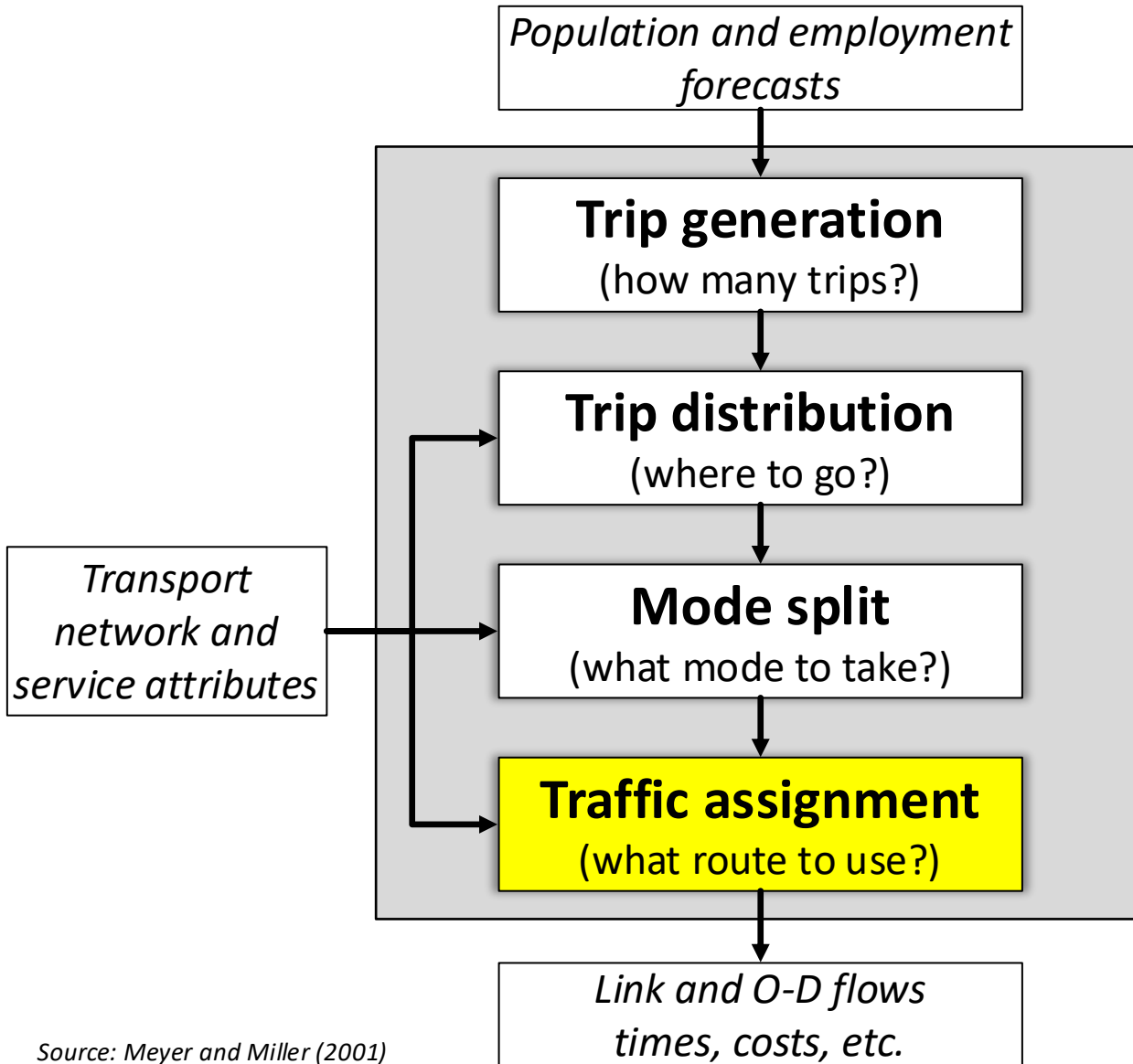
- Key components
 - Path/link flows, Path/link travel times, Link performance function, path assignment rule
- Path Assignment Rules
 - Principle of User Equilibrium (UE)
 - Principle of System Optimisation (SO)
- Static Traffic Assignment (STA) VS. Dynamic Traffic Assignment (DTA)

► References

- Bolyes, Lownes, and Unnikrishnan (BLU) Book
 - Chapter 4. Introduction to Static Assignment
 - Sections 1.5 and 1.6

Traffic Assignment: Key Components

Traffic Assignment in 4-Step Model



- ▶ This course will focus on a road network with only the **automobile mode** (no other modes such as bus, taxi, truck, or bike).
- ▶ We assume that **the travel demand input** (OD matrix) is known to us.
 - OD matrix contains the predicted flows (of the automobile mode) between each origin-destination pair.

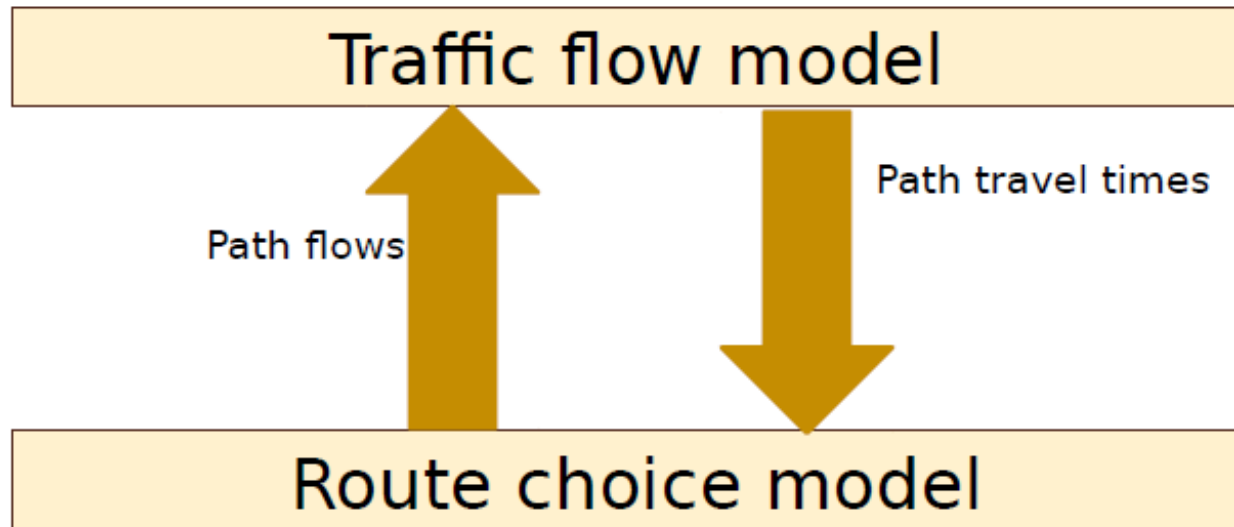
Traffic Assignment Problem

- ▶ Given:
 - A graph representation of the urban transportation network
 - The associated link performance functions
 - An origin-destination (OD) matrix

- ▶ **Find the flow (and the travel time) on each of the network links.**

Traffic Assignment as Interacting Systems

- ▶ Find the **equilibrium solution**: the flows and corresponding travel times at the demand-performance equilibrium (**stable condition**)
- *Given the predicted path flows, the traffic flow model predicts the corresponding path travel times. When these path travel times are used to predict route choices, the route choice model should produce the same path flows (that were initially supplied to the traffic flow model).*



Mathematical Notation (following the BLU book)

- ▶ N : the set of nodes, i.e., node $i \in N$
- ▶ A : the set of links, i.e., link $(i, j) \in A$
- ▶ Z : the set of zone centroids, which is a subset of node set N (i.e., $Z \subseteq N$)
 - r, s : the origin and destination zone centroids, respectively; $r \in Z$ and $s \in Z$
 - (r, s) : the OD-pair between origin r and destination s ; $(r, s) \in Z^2$
- ▶ Π^{rs} : the set of paths between origin r and destination s , i.e., path $\pi \in \Pi^{rs}$
- ▶ Π : the set of all paths, i.e., $\bigcup_{(r,s) \in Z^2} \Pi^{rs}$ (i.e., $\Pi^{rs} \subseteq \Pi$)
- ▶ d^{rs} : trip demand between origin r and destination s ; $\mathbf{d} = (\dots, d^{rs}, \dots)$
- ▶ x_{ij} : link flow on link (i, j) ; $\mathbf{x} = (\dots, x_{ij}, \dots)$
- ▶ t_{ij} : link travel time on link (i, j) ; $\mathbf{t} = (\dots, t_{ij}, \dots)$
- ▶ h^π : path flow on path π ; $\mathbf{h} = (\dots, h^\pi, \dots)$
- ▶ c^π : path travel time on path π ; $\mathbf{c} = (\dots, c^\pi, \dots)$
- ▶ δ_{ij}^π : indicator variable; $\delta_{ij}^\pi = 1$ if link (i, j) is on path π and $\delta_{ij}^\pi = 0$ otherwise
- ▶ Δ : *link-path adjacency matrix* with δ_{ij}^π for link row (i, j) and path column π

Traffic Assignment as an Iterative Process

- **What assignment rule?** How each driver chooses a route, given a set of routes and their predicted travel times?

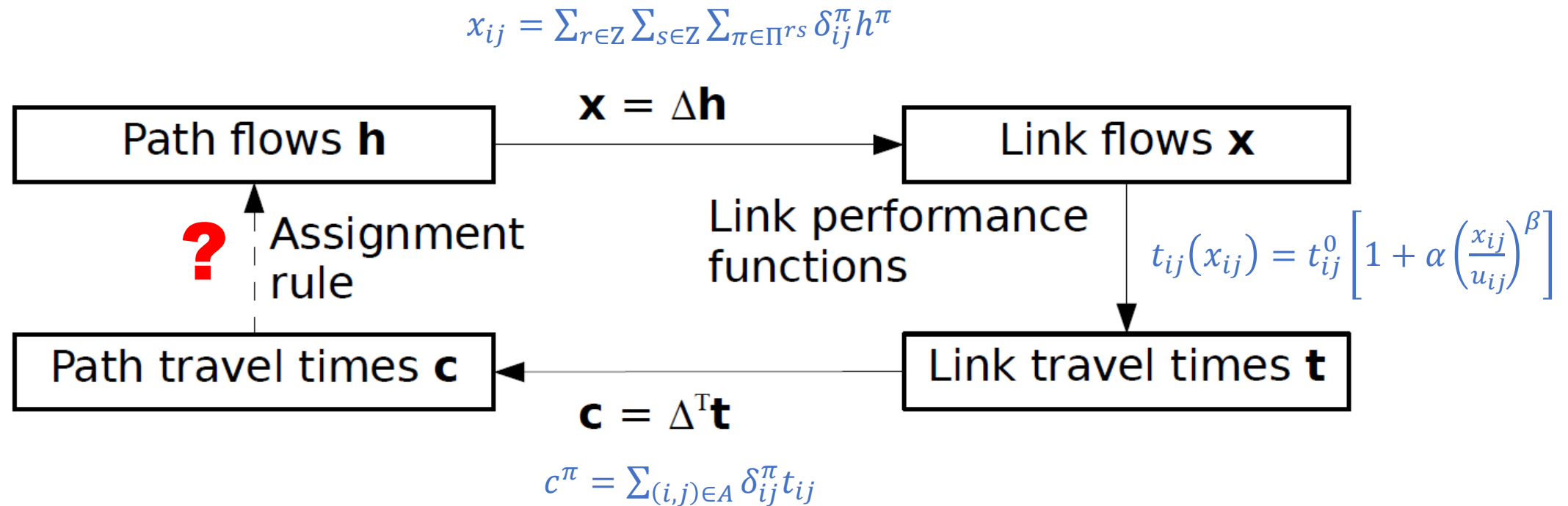
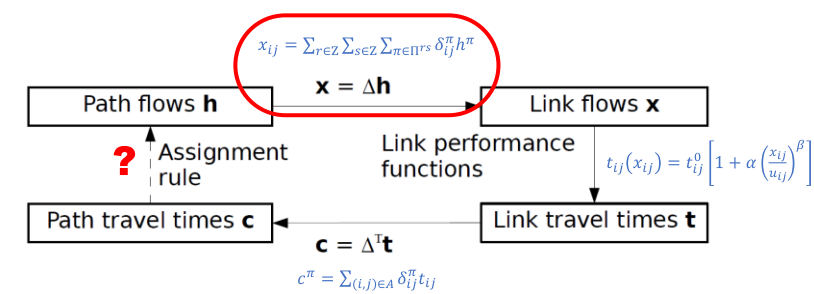


Figure 4.2: Traffic assignment as an iterative process.

From Path Flows $\mathbf{h}$ to Link Flows $\mathbf{x}$

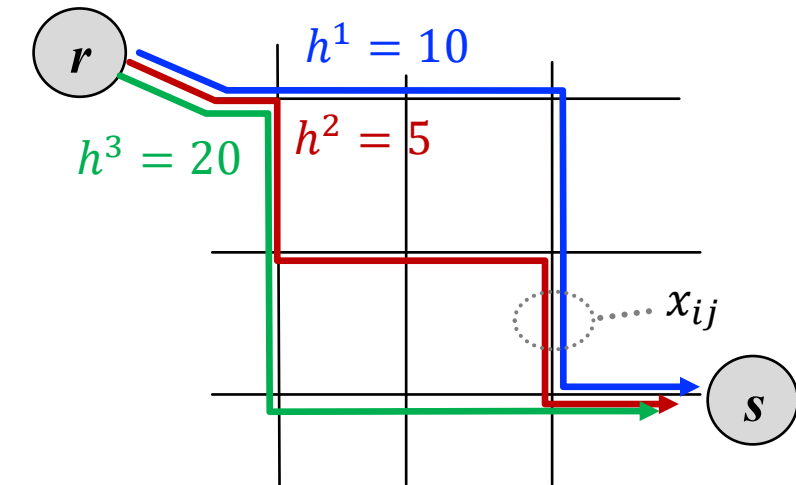


- ▶ Link flows can be determined from path flows.

$$x_{ij} = \sum_{r \in Z} \sum_{s \in Z} \sum_{\pi \in \Pi^{rs}} \delta_{ij}^{\pi} h^{\pi}$$

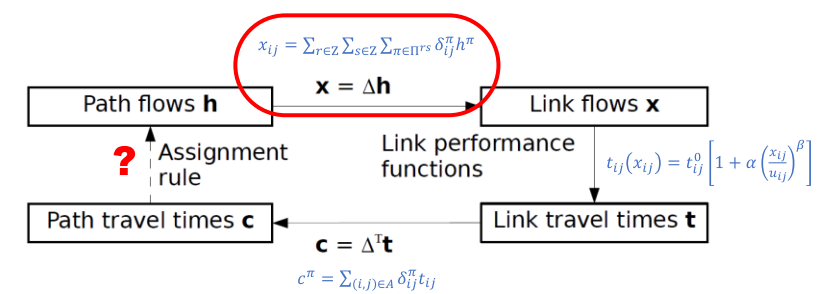
- h^{π} : path flow on path π
- Z is the set of centroids, which is a subset of node set N ($Z \subseteq N$)
- r, s : the origin and destination zone centroids
- Π^{rs} : the set of acyclic paths between origin r and destination s
- δ_{ij}^{π} : indicates whether link (i, j) is traversed by path π :
 - $\delta_{ij}^{\pi} = 0$ if path π does not use link (i, j) and $\delta_{ij}^{\pi} = 1$ if it does.

[Example]



$$\begin{aligned} x_{ij} &= \delta_{ij}^1 h^1 + \delta_{ij}^2 h^2 + \delta_{ij}^3 h^3 \\ &= 1(10) + 1(5) + 0(20) \\ &= 15 \end{aligned}$$

From Path Flows $\mathbf{h}$ to Link Flows $\mathbf{x}$



► Link flows can be determined from path flows.

$$x_{ij} = \sum_{r \in Z} \sum_{s \in Z} \sum_{\pi \in \Pi^{rs}} \delta_{ij}^{\pi} h^{\pi}$$

► [Example]

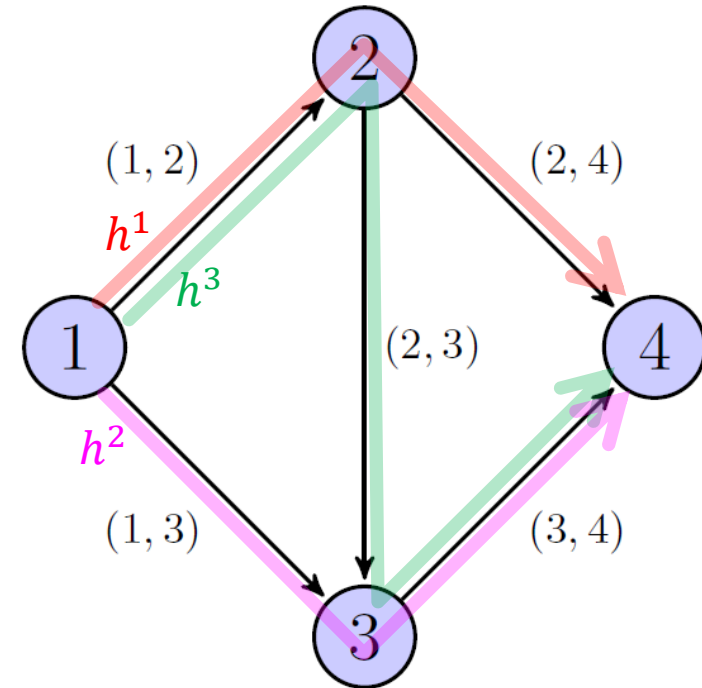
$$x_{12} = \delta_{12}^1 h^1 + \delta_{12}^2 h^2 + \delta_{12}^3 h^3$$

$$x_{13} = \delta_{13}^1 h^1 + \delta_{13}^2 h^2 + \delta_{13}^3 h^3$$

$$x_{23} = \delta_{23}^1 h^1 + \delta_{23}^2 h^2 + \delta_{23}^3 h^3$$

$$x_{24} = \delta_{24}^1 h^1 + \delta_{24}^2 h^2 + \delta_{24}^3 h^3$$

$$x_{34} = \delta_{34}^1 h^1 + \delta_{34}^2 h^2 + \delta_{34}^3 h^3$$

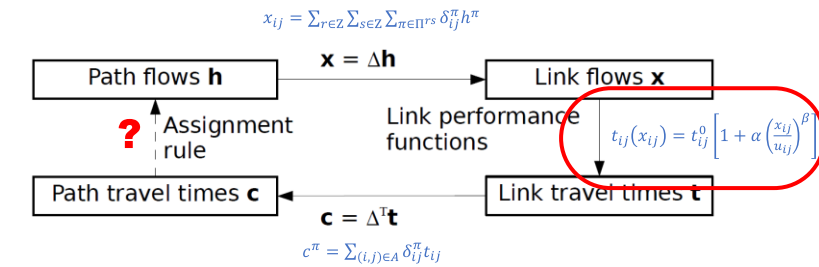


$$\pi_1 = [1, 2, 4], \quad h^1 = 10$$

$$\pi_2 = [1, 3, 4], \quad h^2 = 5$$

$$\pi_3 = [1, 2, 3, 4], \quad h^3 = 20$$

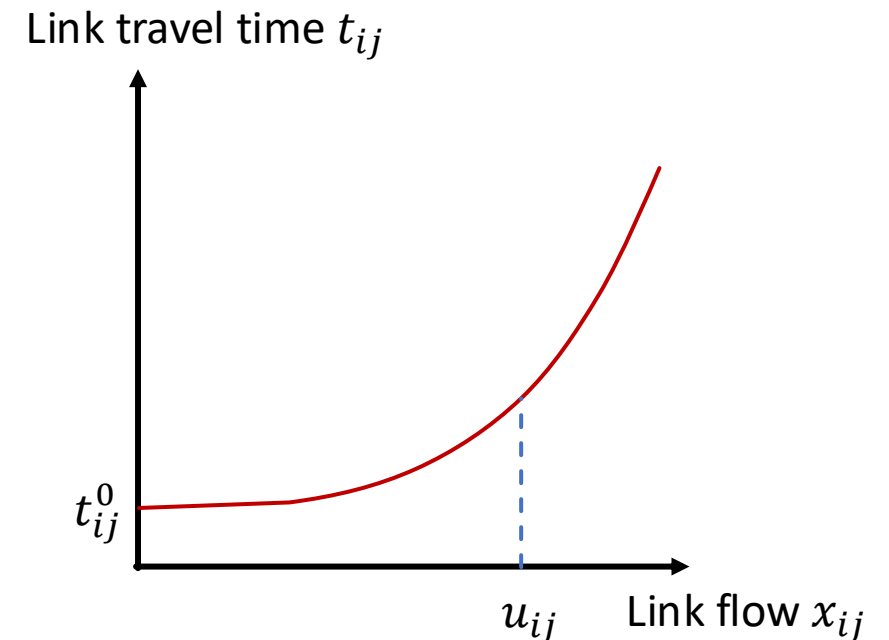
From Link Flows $\mathbf{x}$ to Link Travel Times $\mathbf{t}$



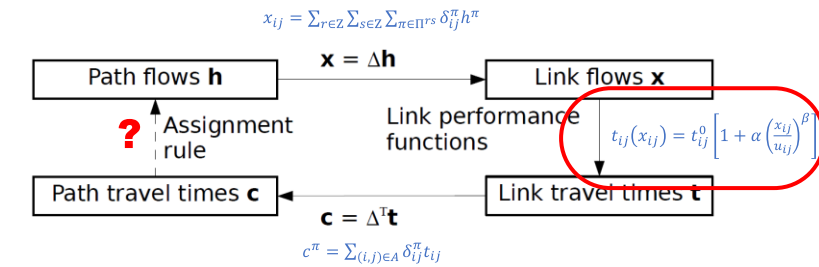
- ▶ Link travel times can be determined using a **link performance function**, which predicts link travel time as a function of link flows.
- ▶ The most popular link performance function is the BPR function, originally proposed by engineers at the U.S. Bureau of Public Roads in 1960.

$$t_{ij}(x_{ij}) = t_{ij}^0 \left[1 + \alpha \left(\frac{x_{ij}}{u_{ij}} \right)^\beta \right]$$

- ▶ t_{ij}^0 : the free-flow travel time on link (i, j)
- ▶ u_{ij} : the capacity of the link
- ▶ α and β : model parameters that can be calibrated to data
($\alpha = 0.15$ and $\beta = 4$ are commonly used)



From Link Flows $\mathbf{x}$ to Link Travel Times $\mathbf{t}$

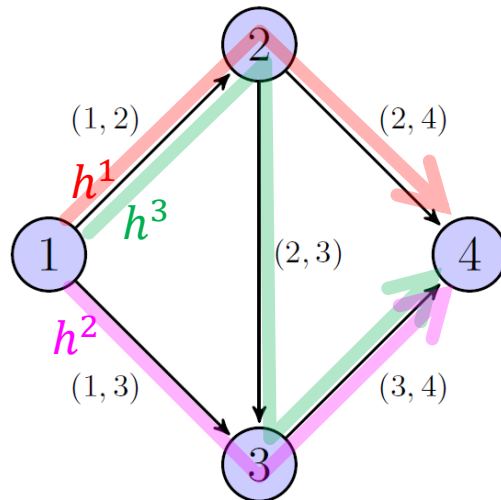


- Link travel times can be determined using a **link performance function**, which predicts link travel time as a function of link flows.

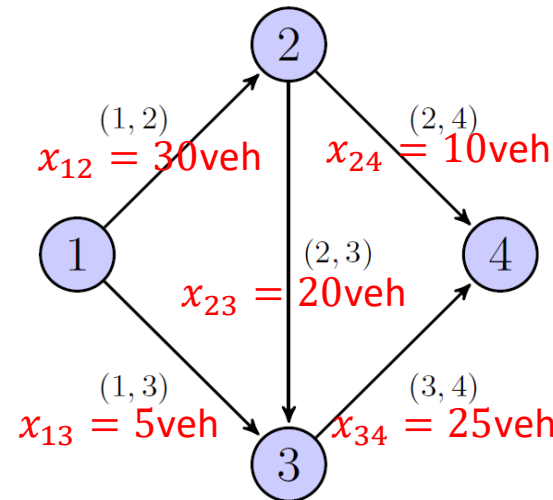
$$t_{ij}(x_{ij}) = t_{ij}^0 \left[1 + \alpha \left(\frac{x_{ij}}{u_{ij}} \right)^\beta \right]$$

- [Example]

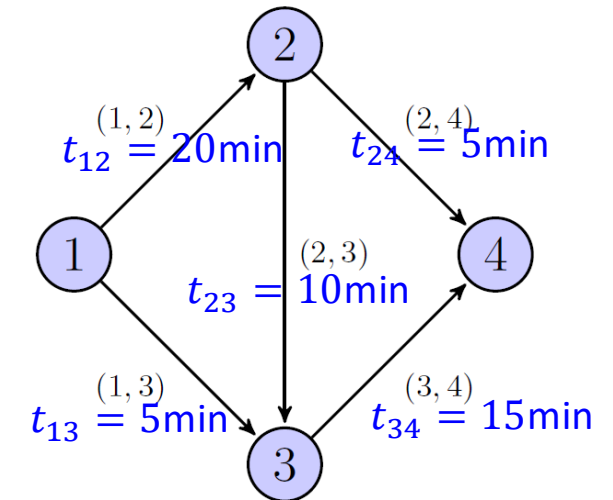
Path assignment & path flows



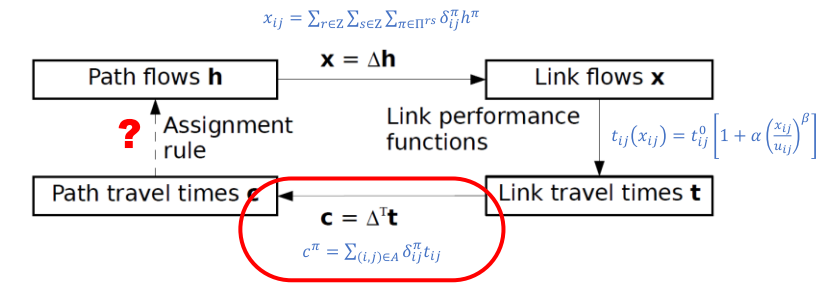
Link flows



Link travel times



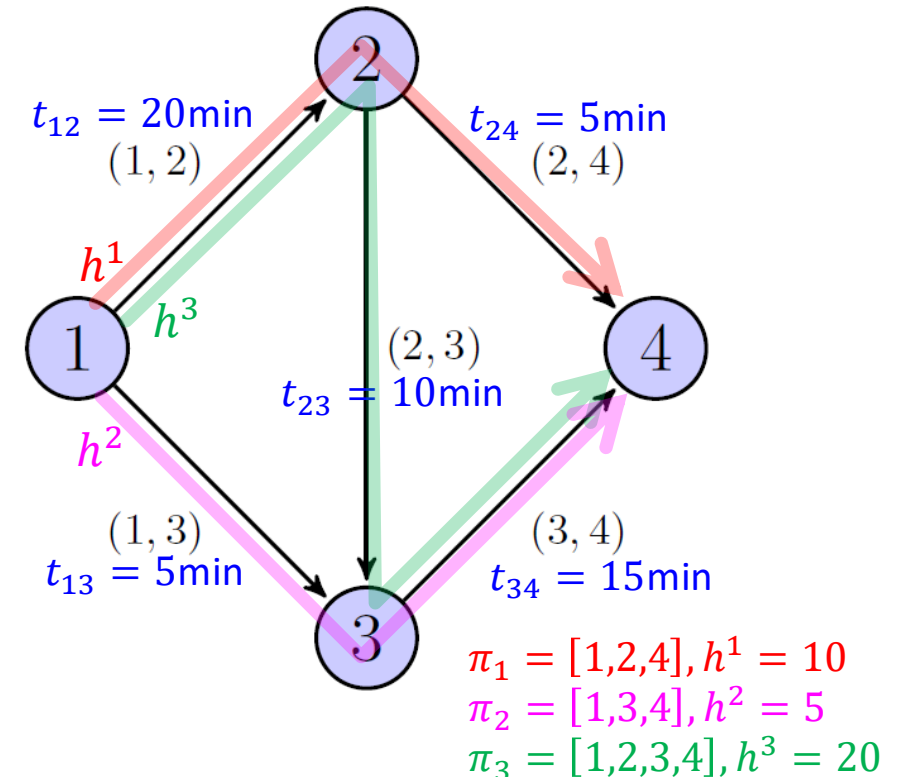
From Link Travel Times $\mathbf{t}$ to Path Travel Times $\mathbf{c}$



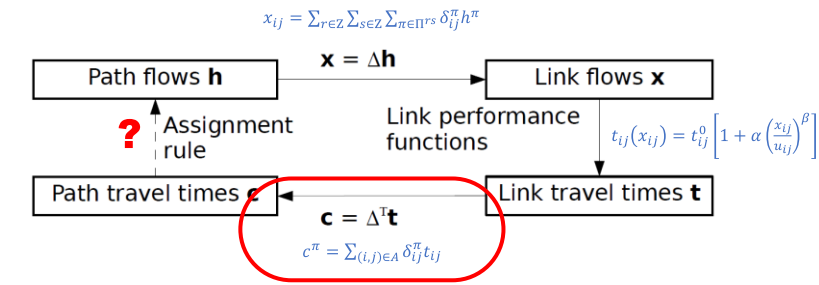
- ▶ The path travel times are directly related to the link travel times.
- ▶ The travel time of a path is the sum of the travel times of the links comprising that path.

$$c^\pi = \sum_{(i,j) \in A} \delta_{ij}^\pi t_{ij}$$

- ▶ c^π : travel time of path π
- ▶ δ_{ij}^π : link-path relationship indicator
- ▶ [Example] The path travel time for $\pi_1 = [1, 2, 4]$ is:
 - ▶ $c^\pi = \delta_{12}^\pi t_{12} + \delta_{13}^\pi t_{13} + \delta_{23}^\pi t_{23} + \delta_{24}^\pi t_{24} + \delta_{34}^\pi t_{34}$
 $= 1(20) + 0(5) + 0(10) + 1(5) + 0(15) = 25$



From Link Travel Times $\mathbf{t}$ to Path Travel Times $\mathbf{c}$



- ▶ The travel time of a path is the sum of the travel times of the links comprising that path.

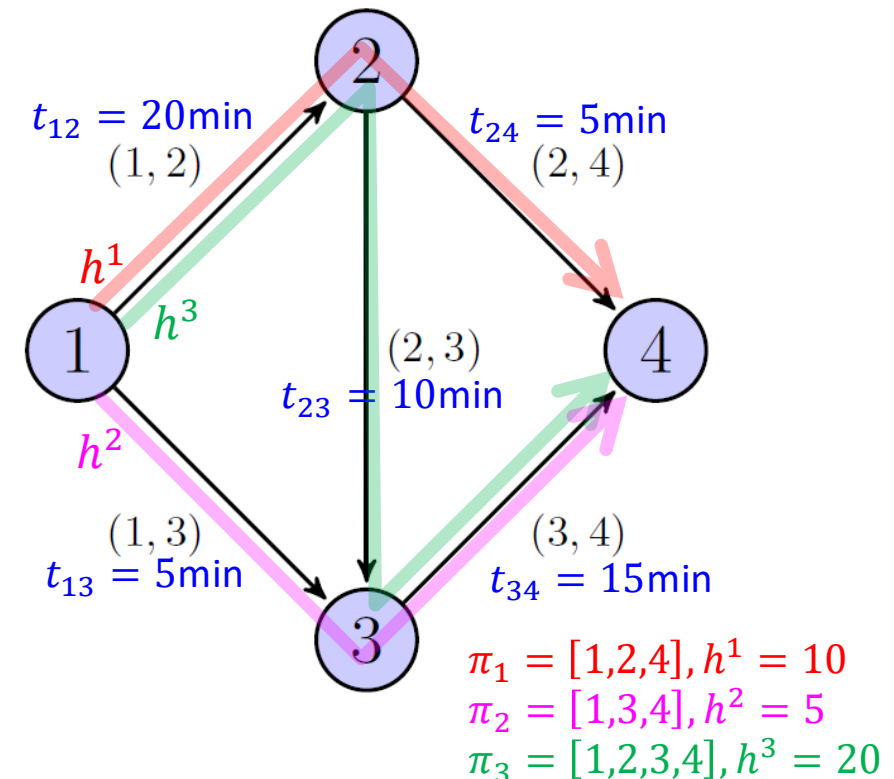
$$c^\pi = \sum_{(i,j) \in A} \delta_{ij}^\pi t_{ij}$$

- ▶ [Example] Path travel times for all paths are:

$$c^1 = \delta_{12}^1 t_{12} + \delta_{13}^1 t_{13} + \delta_{23}^1 t_{23} + \delta_{24}^1 t_{24} + \delta_{34}^1 t_{34}$$

$$c^2 = \delta_{12}^2 t_{12} + \delta_{13}^2 t_{13} + \delta_{23}^2 t_{23} + \delta_{24}^2 t_{24} + \delta_{34}^2 t_{34}$$

$$c^3 = \delta_{12}^3 t_{12} + \delta_{13}^3 t_{13} + \delta_{23}^3 t_{23} + \delta_{24}^3 t_{24} + \delta_{34}^3 t_{34}$$



Link-Path Adjacency Matrix, Δ

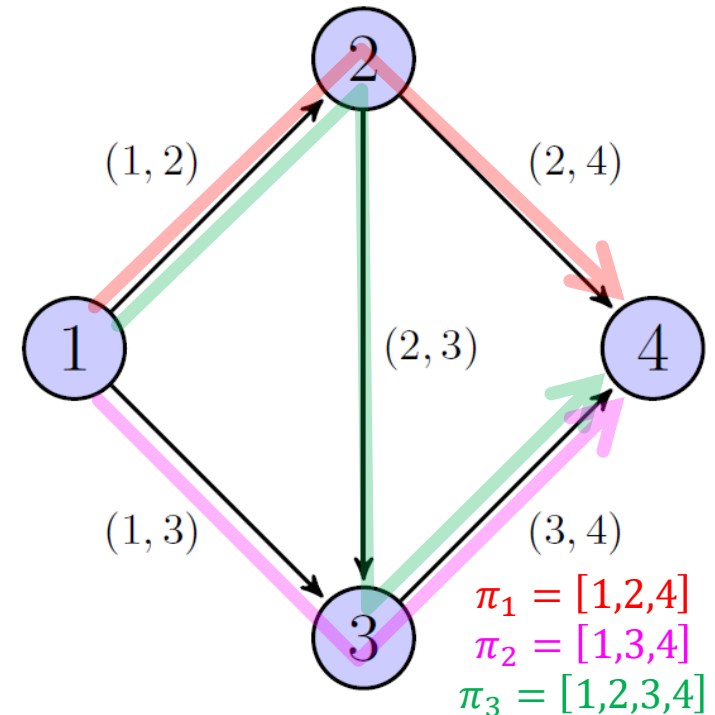
► A matrix of δ_{ij}^π , where **rows** represent **links** (i, j) and **columns** represent **paths** π . The elements δ_{ij}^π indicate **whether a particular link is traversed by a particular path**.

► $\delta_{ij}^\pi = 1$ indicates that link (i, j) is part of path π and $\delta_{ij}^\pi = 0$ indicates it is not.

▪ Links in rows are **sorted first by upstream node ID in ascending order, and then by downstream node ID in ascending order**.

► [Example] Consider three paths between OD-pair $1 \rightarrow 4$:

► $\Pi^{14} = \{\pi_1, \pi_2, \pi_3\} = \{[1,2,4], [1,3,4], [1,2,3,4]\}$



Link \ Path	π_1	π_2	π_3
(1,2)	1	0	1
(1,3)	0	1	0
(2,3)	0	0	1
(2,4)	1	0	0
(3,4)	0	1	1

Link-path
adjacency
matrix $\Delta =$

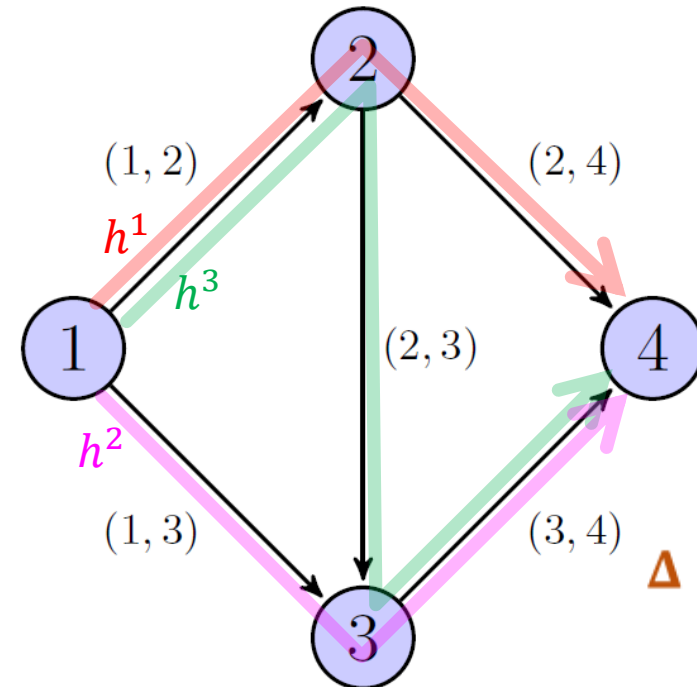
Matrix Representation $\mathbf{x} = \Delta \mathbf{h}$

: A Mapping from Path Flows $\mathbf{h}$ to Link Flows $\mathbf{x}$

- The relation $x_{ij} = \sum_{r \in Z} \sum_{s \in Z} \sum_{\pi \in \Pi^{rs}} \delta_{ij}^{\pi} h^{\pi}$ for all links can be compactly written as $\mathbf{x} = \Delta \mathbf{h}$, where $\mathbf{x}$ and $\mathbf{h}$ are *column vectors* and Δ is a *matrix*.

$$\mathbf{x} = \Delta \cdot \mathbf{h}$$

$$\begin{bmatrix} x_{12} \\ x_{13} \\ x_{23} \\ x_{24} \\ x_{34} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} h^1 \\ h^2 \\ h^3 \end{bmatrix} = \begin{bmatrix} h^1 + h^3 \\ h^2 \\ h^3 \\ h^1 \\ h^2 + h^3 \end{bmatrix}$$



$$\begin{aligned} \pi_1 &= [1,2,4] \\ \pi_2 &= [1,3,4] \\ \pi_3 &= [1,2,3,4] \end{aligned}$$

$\Delta =$

Path Link	π_1	π_2	π_3
(1,2)	1	0	1
(1,3)	0	1	0
(2,3)	0	0	1
(2,4)	1	0	0
(3,4)	0	1	1

Operator ' $\cdot$ ' denotes matrix multiplication.

Matrix Representation $\mathbf{x} = \Delta \mathbf{h}$

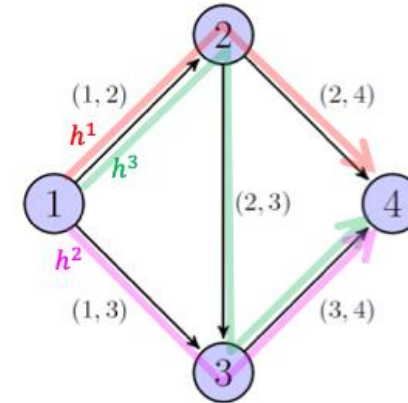
: A Mapping from Path Flows $\mathbf{h}$ to Link Flows $\mathbf{x}$

► The relation $x_{ij} = \sum_{r \in Z} \sum_{s \in Z} \sum_{\pi \in \Pi^{rs}} \delta_{ij}^{\pi} h^{\pi}$ for all links can be compactly written as $\mathbf{x} = \Delta \mathbf{h}$.

$$\mathbf{x} = \Delta \cdot \mathbf{h}$$

$$\begin{bmatrix} x_{12} \\ x_{13} \\ x_{23} \\ x_{24} \\ x_{34} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} h^1 \\ h^2 \\ h^3 \end{bmatrix} = \begin{bmatrix} h^1 + h^3 \\ h^2 \\ h^3 \\ h^1 \\ h^2 + h^3 \end{bmatrix}$$

Operator ' $\cdot$ ' denotes matrix multiplication.



$$x_{12} = \delta_{12}^1 h^1 + \delta_{12}^2 h^2 + \delta_{12}^3 h^3 = 1(10) + 0(5) + 1(20) = 30$$

$$x_{13} = \delta_{13}^1 h^1 + \delta_{13}^2 h^2 + \delta_{13}^3 h^3 = 0(10) + 1(5) + 0(20) = 5$$

$$x_{23} = \delta_{23}^1 h^1 + \delta_{23}^2 h^2 + \delta_{23}^3 h^3 = 0(10) + 0(5) + 1(20) = 20$$

$$x_{24} = \delta_{24}^1 h^1 + \delta_{24}^2 h^2 + \delta_{24}^3 h^3 = 1(10) + 0(5) + 0(20) = 10$$

$$x_{34} = \delta_{34}^1 h^1 + \delta_{34}^2 h^2 + \delta_{34}^3 h^3 = 0(10) + 1(5) + 1(20) = 25$$

Matrix Representation $\mathbf{c} = \Delta^T \mathbf{t}$

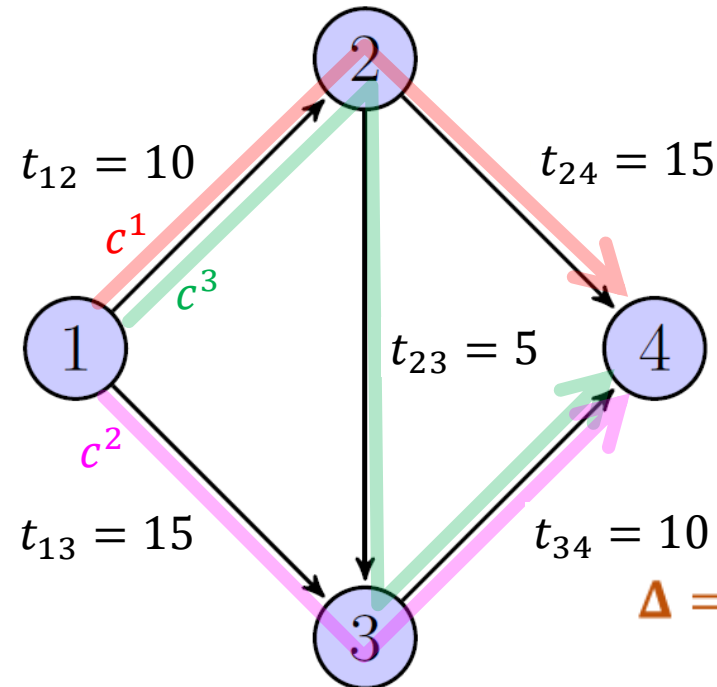
: A Mapping from Link Travel Times $\mathbf{t}$ to Path Travel Times $\mathbf{c}$

- The relation $c^\pi = \sum_{(i,j) \in A} \delta_{ij}^\pi t_{ij}$ for all paths can be compactly written as $\mathbf{c} = \Delta^T \mathbf{t}$, where $\mathbf{c}$ and $\mathbf{t}$ are *column vectors* and Δ^T (the transpose of Δ) is a *matrix*.

$$\mathbf{c} = \Delta^T \cdot \mathbf{t}$$

$$\begin{bmatrix} c^1 \\ c^2 \\ c^3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} t_{12} \\ t_{13} \\ t_{23} \\ t_{24} \\ t_{34} \end{bmatrix}$$

$$= \begin{bmatrix} t_{12} + t_{24} \\ t_{13} + t_{34} \\ t_{12} + t_{23} + t_{34} \end{bmatrix}$$



$$\begin{aligned} \pi_1 &= [1,2,4] \\ \pi_2 &= [1,3,4] \\ \pi_3 &= [1,2,3,4] \end{aligned}$$

$$\Delta =$$

Path Link	π_1	π_2	π_3
(1,2)	1	0	1
(1,3)	0	1	0
(2,3)	0	0	1
(2,4)	1	0	0
(3,4)	0	1	1

Matrix Representation $\mathbf{c} = \Delta^T \mathbf{t}$

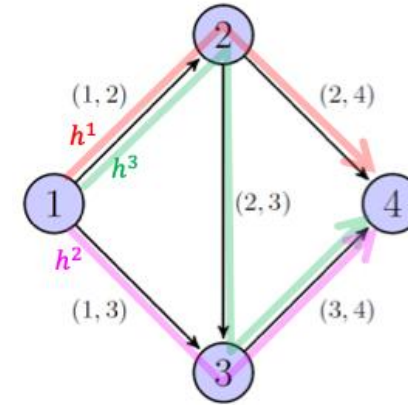
: A Mapping from Link Travel Times $\mathbf{t}$ to Path Travel Times $\mathbf{c}$

► The relation $c^\pi = \sum_{(i,j) \in A} \delta_{ij}^\pi t_{ij}$ for all paths can be compactly written as $\mathbf{c} = \Delta^T \mathbf{t}$.

$$\mathbf{c} = \Delta^T \cdot \mathbf{t}$$

$$\begin{bmatrix} c^1 \\ c^2 \\ c^3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} t_{12} \\ t_{13} \\ t_{23} \\ t_{24} \\ t_{34} \end{bmatrix}$$

$$= \begin{bmatrix} t_{12} + t_{24} \\ t_{13} + t_{34} \\ t_{12} + t_{23} + t_{34} \end{bmatrix}$$



$$\begin{aligned} c^1 &= \delta_{12}^1 t_{12} + \delta_{13}^1 t_{13} + \delta_{23}^1 t_{23} + \delta_{24}^1 t_{24} + \delta_{34}^1 t_{34} \\ &= 1(20) + 0(5) + 0(10) + 1(5) + 0(15) = 25 \end{aligned}$$

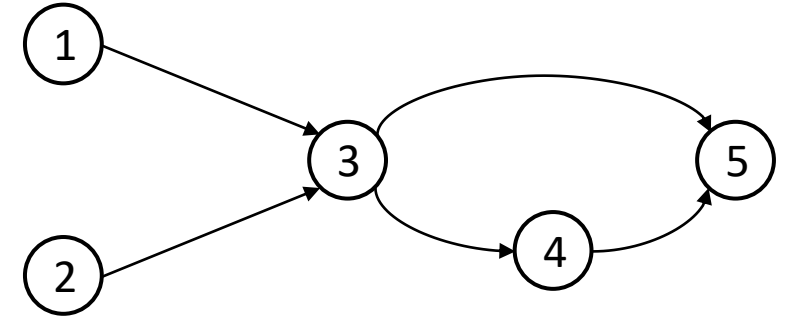
$$\begin{aligned} c^2 &= \delta_{12}^2 t_{12} + \delta_{13}^2 t_{13} + \delta_{23}^2 t_{23} + \delta_{24}^2 t_{24} + \delta_{34}^2 t_{34} \\ &= 0(20) + 1(5) + 0(10) + 0(5) + 1(15) = 20 \end{aligned}$$

$$\begin{aligned} c^3 &= \delta_{12}^3 t_{12} + \delta_{13}^3 t_{13} + \delta_{23}^3 t_{23} + \delta_{24}^3 t_{24} + \delta_{34}^3 t_{34} \\ &= 1(20) + 0(5) + 1(10) + 0(5) + 1(15) = 45 \end{aligned}$$

Exercise

Consider four paths between two OD-pairs $1 \rightarrow 5$ and $2 \rightarrow 5$ in the network on the right:

$$\pi_1 = [1, 3, 5], \quad \pi_2 = [1, 3, 4, 5], \quad \pi_3 = [2, 3, 5], \quad \pi_4 = [2, 3, 4, 5].$$



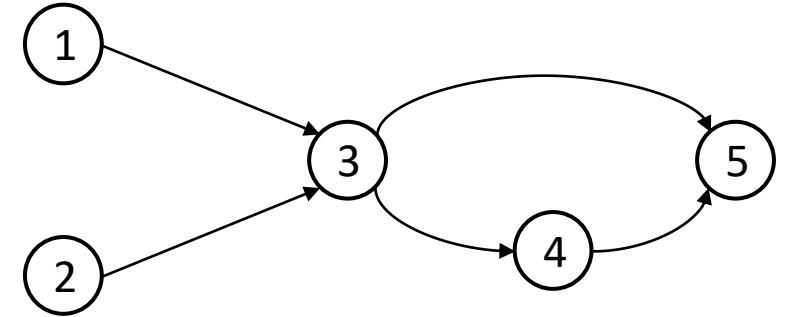
- ▶ Write the link-path adjacency matrix, Δ .
- ▶ Write the relationship between link flow $\mathbf{x} = (\dots, x_{ij}, \dots)$ and path flows $\mathbf{h} = (\dots, h^\pi, \dots)$ using matrix notation. Show how each link flow x_{ij} is expressed in terms of path flows h^π .
- ▶ Write the relationship between link travel times $\mathbf{t} = (\dots, t_{ij}, \dots)$ and path travel times $\mathbf{c} = (\dots, c^\pi, \dots)$ using matrix notation. Show how each path travel time c^π is expressed in terms of link travel times t_{ij} .

Exercise

Consider four paths between two OD-pairs $1 \rightarrow 5$ and $2 \rightarrow 5$ in the network on the right:

$\pi_1 = [1, 3, 5]$, $\pi_2 = [1, 3, 4, 5]$, $\pi_3 = [2, 3, 5]$, $\pi_4 = [2, 3, 4, 5]$.

► Write the link-path adjacency matrix, Δ .



Link-path
adjacency
matrix

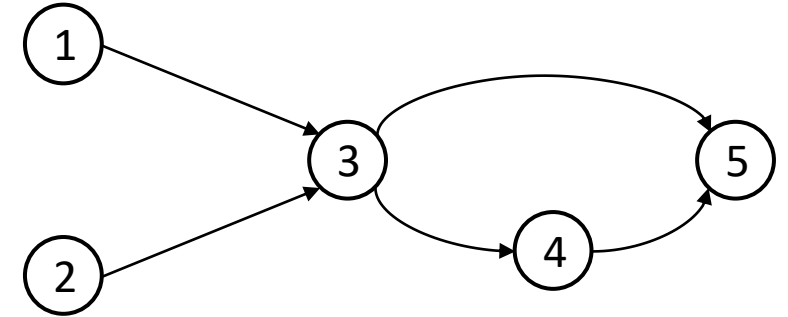
$\Delta =$

Path Link	π_1	π_2	π_3	π_4
(1,3)				
(2,3)				
(3,4)				
(3,5)				
(4,5)	-	-	-	-

Exercise

Consider four paths between two OD-pairs $1 \rightarrow 5$ and $2 \rightarrow 5$ in the network on the right:

$$\pi_1 = [1, 3, 5], \quad \pi_2 = [1, 3, 4, 5], \quad \pi_3 = [2, 3, 5], \quad \pi_4 = [2, 3, 4, 5].$$



► Write the link-path adjacency matrix, Δ .

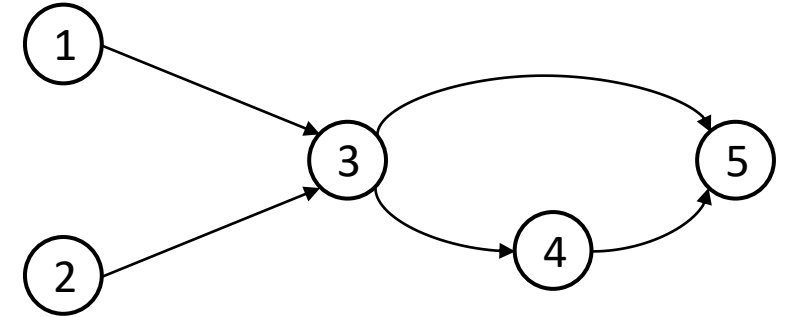
Link-path adjacency matrix $\Delta =$

Path Link	π_1	π_2	π_3	π_4
(1,3)	1	1	0	0
(2,3)	0	0	1	1
(3,4)	0	1	0	1
(3,5)	1	0	1	0
(4,5)	0	1	0	1

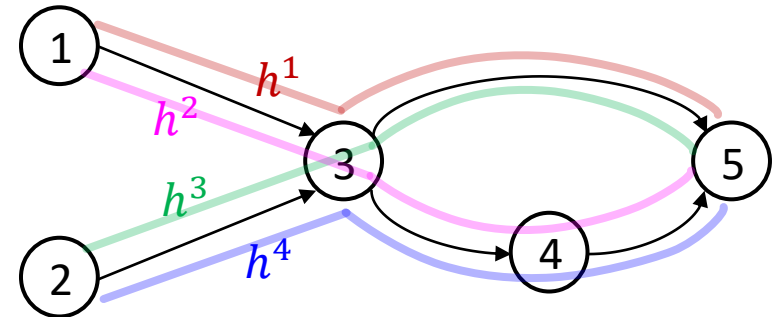
Exercise

Consider four paths between two OD-pairs $1 \rightarrow 5$ and $2 \rightarrow 5$ in the network on the right:

$$\pi_1 = [1, 3, 5], \quad \pi_2 = [1, 3, 4, 5], \quad \pi_3 = [2, 3, 5], \quad \pi_4 = [2, 3, 4, 5].$$



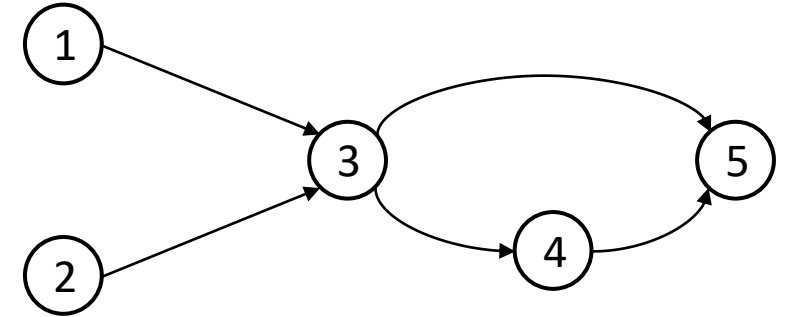
- Write the relationship between link flow $\mathbf{x} = (\dots, x_{ij}, \dots)$ and path flows $\mathbf{h} = (\dots, h^\pi, \dots)$ using matrix notation. Show how each link flow x_{ij} is expressed in terms of path flows h^π .



Exercise

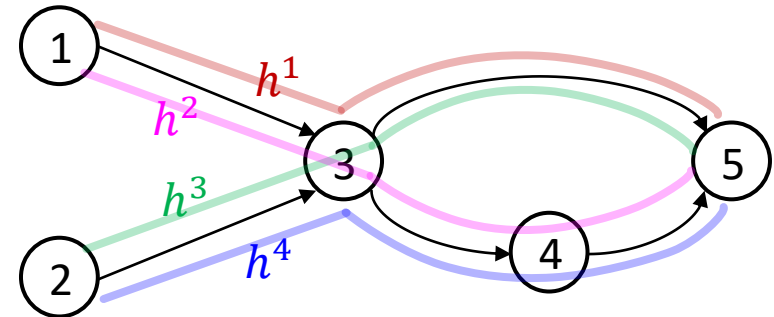
Consider four paths between two OD-pairs $1 \rightarrow 5$ and $2 \rightarrow 5$ in the network on the right:

$$\pi_1 = [1, 3, 5], \quad \pi_2 = [1, 3, 4, 5], \quad \pi_3 = [2, 3, 5], \quad \pi_4 = [2, 3, 4, 5].$$



- Write the relationship between link flow $\mathbf{x} = (\dots, x_{ij}, \dots)$ and path flows $\mathbf{h} = (\dots, h^\pi, \dots)$ using matrix notation. Show how each link flow x_{ij} is expressed in terms of path flows h^π .

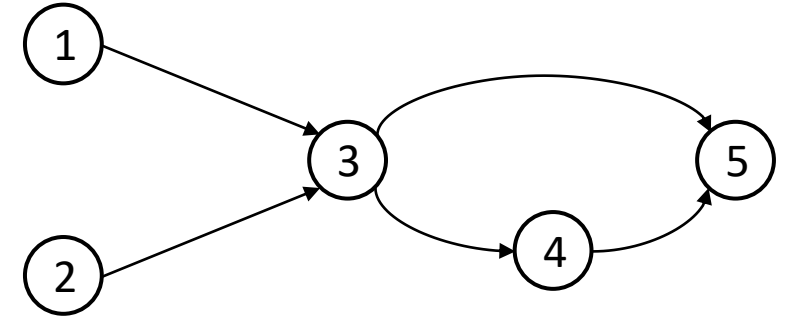
$$\mathbf{x} = \Delta \mathbf{h}$$
$$\begin{bmatrix} x_{13} \\ x_{23} \\ x_{34} \\ x_{35} \\ x_{45} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} h^1 \\ h^2 \\ h^3 \\ h^4 \end{bmatrix} = \begin{bmatrix} h^1 + h^2 \\ h^3 + h^4 \\ h^2 + h^4 \\ h^1 + h^3 \\ h^2 + h^4 \end{bmatrix}$$



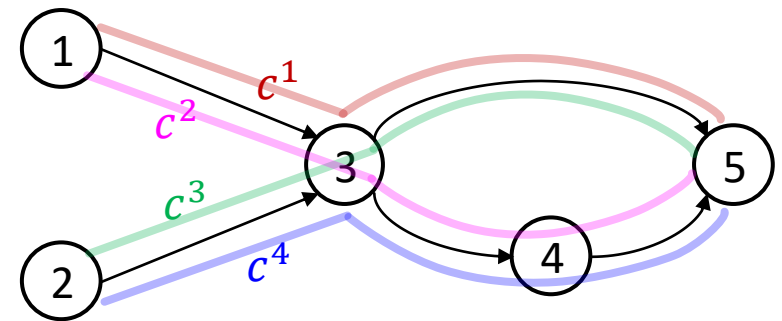
Exercise

Consider four paths between two OD-pairs $1 \rightarrow 5$ and $2 \rightarrow 5$ in the network on the right:

$$\pi_1 = [1, 3, 5], \quad \pi_2 = [1, 3, 4, 5], \quad \pi_3 = [2, 3, 5], \quad \pi_4 = [2, 3, 4, 5].$$



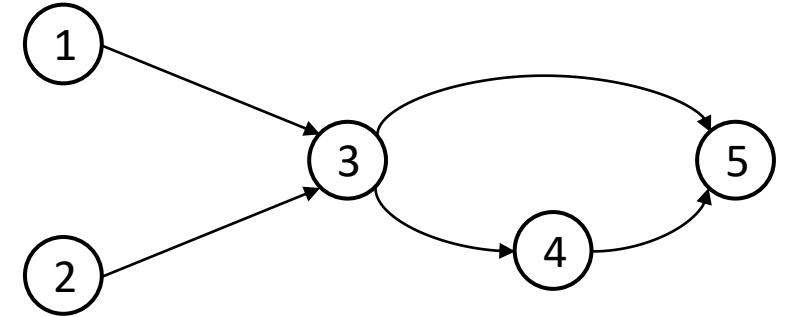
- Write the relationship between link travel times $\mathbf{t} = (\dots, t_{ij}, \dots)$ and path travel times $\mathbf{c} = (\dots, c^\pi, \dots)$ using matrix notation. Show how each path travel time c^π is expressed in terms of link travel times t_{ij} .



Exercise

Consider four paths between two OD-pairs $1 \rightarrow 5$ and $2 \rightarrow 5$ in the network on the right:

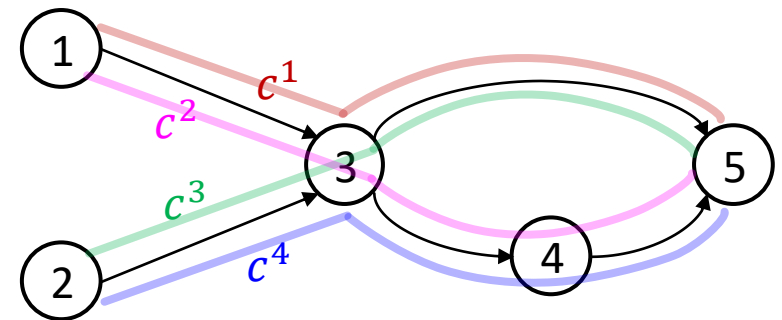
$$\pi_1 = [1, 3, 5], \quad \pi_2 = [1, 3, 4, 5], \quad \pi_3 = [2, 3, 5], \quad \pi_4 = [2, 3, 4, 5].$$



- Write the relationship between link travel times $\mathbf{t} = (\dots, t_{ij}, \dots)$ and path travel times $\mathbf{c} = (\dots, c^\pi, \dots)$ using matrix notation. Show how each path travel time c^π is expressed in terms of link travel times t_{ij} .

$$\mathbf{c} = \Delta^T \mathbf{t}$$

$$\begin{bmatrix} c^1 \\ c^2 \\ c^3 \\ c^4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} t_{13} \\ t_{23} \\ t_{34} \\ t_{35} \\ t_{45} \end{bmatrix} = \begin{bmatrix} t_{13} + t_{35} \\ t_{13} + t_{34} + t_{45} \\ t_{23} + t_{35} \\ t_{23} + t_{34} + t_{45} \end{bmatrix}$$



Traffic Assignment as an Iterative Process

- **What assignment rule?** How each driver chooses his/her route, given a set of routes and their predicted travel times?

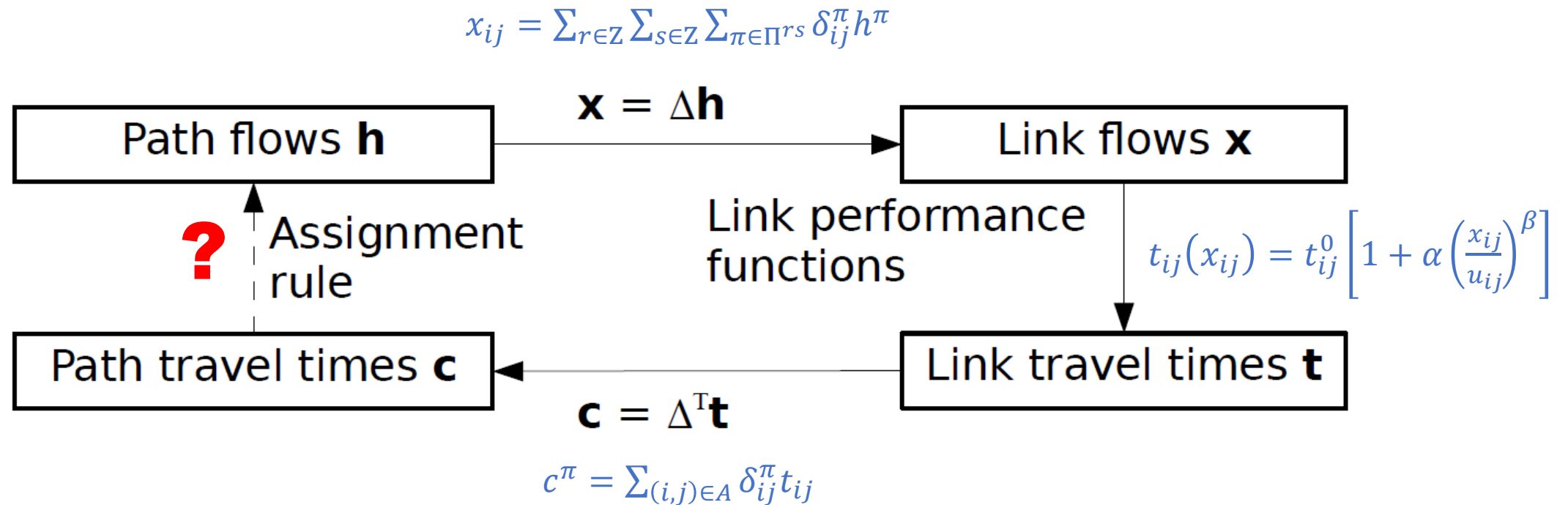


Figure 4.2: Traffic assignment as an iterative process.

User Equilibrium (UE)

Route Assignment Rule

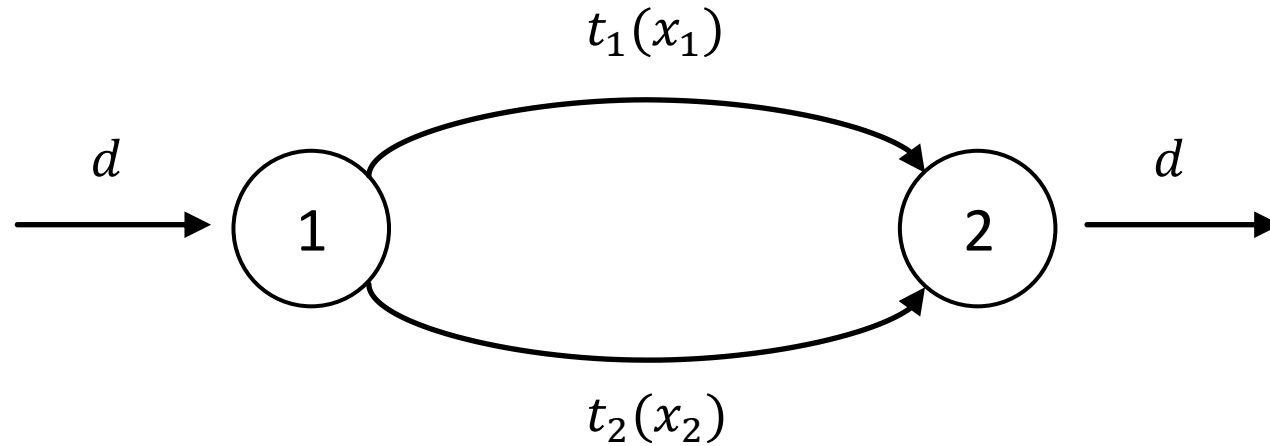
User Equilibrium (UE) Assignment Rule

- ▶ A **stable condition** is reached only when '*no traveller can improve their travel time by unilaterally changing routes*'.
- ▶ This is the principle of **User-Equilibrium** (UE).
- ▶ At UE, '*all used paths between the same origin and destination have **equal and minimal travel times***'.
 - Because, if this were not the case, travellers would switch from slower routes to faster ones, which would tend to equalise their travel times.

Basic Assumptions for User Equilibrium

- ▶ **Assumption 1 (Shortest Path Assumption):** *Each driver wants to choose the path between their origin and their destination with the least travel time.*
- ▶ **Assumption 2 (Perfect Information Assumption):** *Drivers have perfect knowledge of link travel times.*
- ▶ **Corollary (Principle of User Equilibrium):** *Every used route connecting an origin and destination has equal and minimal travel time.*

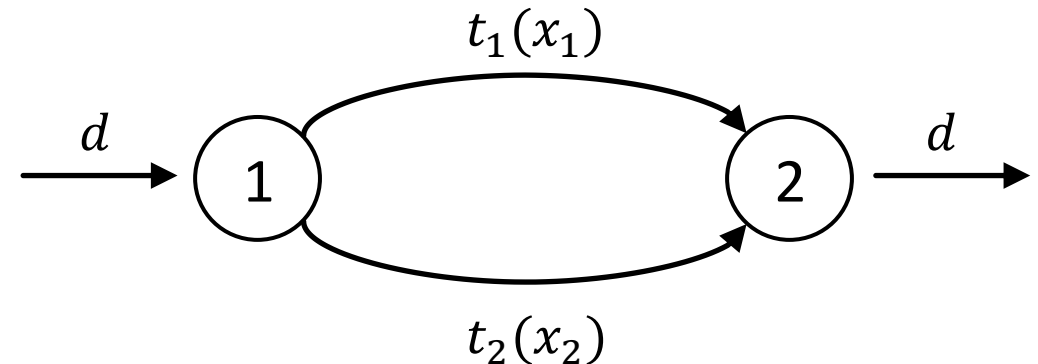
Example 1



- ▶ Consider a network with two paths connecting zone 1 and zone 2.
- ▶ Each path consists of a single link.
 - Link flows = path flows ($x_1 = h_1, x_2 = h_2$)
 - Link travel times = path travel times ($c_1 = t_1, c_2 = t_2$)
- ▶ Travel demand from zone 1 to zone 2 is d .

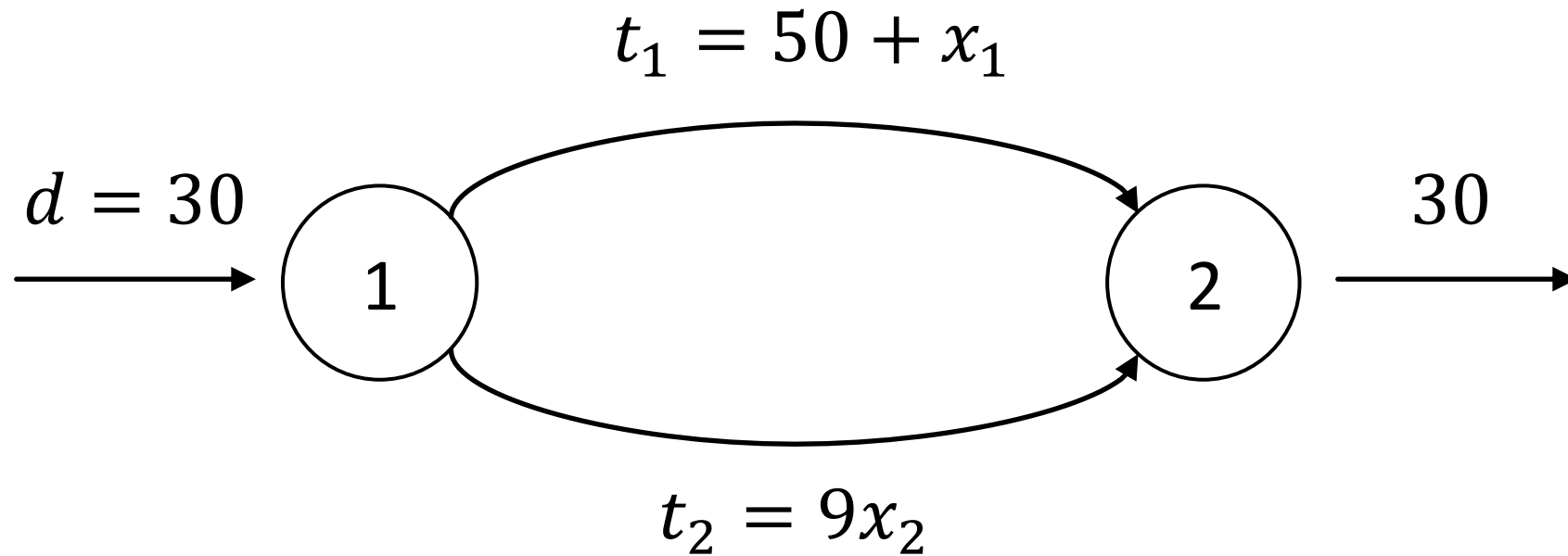
Example 1

- ▶ If all drivers are choosing routes that minimize their travel time **under the UE condition**, then exactly one of the following cases must be true:
 - All drivers are on Path 1 (top link): $t_1(d) < t_2(0)$
 - Path 1 is faster, even when all d drivers are using it
 - All drivers are on Path 2 (bottom link): $t_2(d) < t_1(0)$
 - Path 2 is faster, even when all d drivers are using it
 - The two paths have equal travel time: $t_1(x_1) = t_2(x_2)$, $x_1 + x_2 = d$
 - Neither path dominates the other and people use both paths, where their travel times are equal.



Example 2

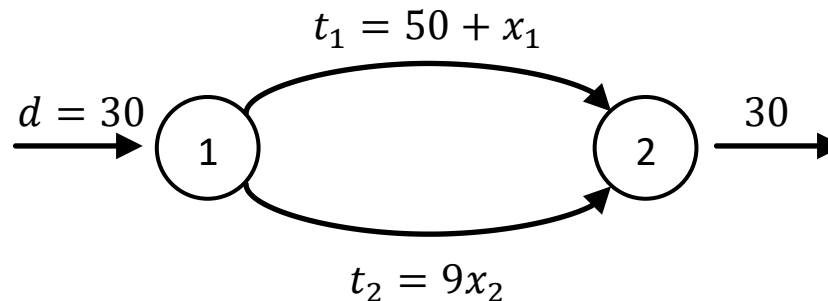
- If 30 vehicles are travelling from zone 1 to zone 2, what is the UE path flows?



Example 2

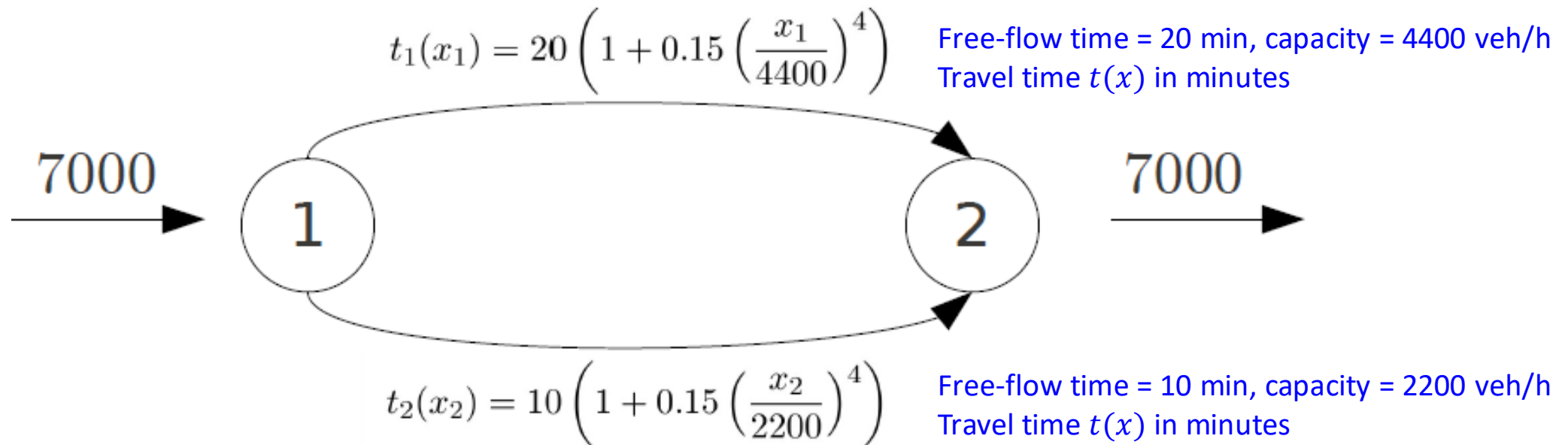
- ▶ Check if any path dominates the other:
 - If all drivers are on Path 1: $t_1(30) = 80 > t_2(0) = 0 \rightarrow$ drivers will switch from Path 1 to Path 2
 - If all drivers are on Path 2: $t_1(0) = 50 < t_2(30) = 270 \rightarrow$ drivers will switch from Path 2 to Path 1
- ▶ Both paths are used and their travel times should be equal:

$$\begin{aligned} t_1(x_1) = t_2(x_2), \quad x_1 + x_2 = 30 \quad \rightarrow \quad & 50 + x_1 = 9x_2 \\ & 50 + 30 - x_2 = 9x_2 \\ & 10x_2 = 80 \\ & x_2 = 8, x_1 = 22 \quad \leftarrow \text{UE path flows} \\ & t_2 = 72, t_1 = 72 \quad \leftarrow \text{UE path travel times} \end{aligned}$$



Example 3

- ▶ A similar approach can be applied even with a more complex link performance function (i.e., BPR function: $t_{ij}(x_{ij}) = t_{ij}^0 \left[1 + \alpha \left(\frac{x_{ij}}{u_{ij}} \right)^\beta \right]$).
 - Solve $t_1(x_1) = t_2(x_2)$ subject to $x_1 + x_2 = 7000$.
 - Try it yourself. You will need an equation solver.



Example 3

► The UE solution:

$$t_1(x_1) = t_2(7000 - x_1)$$

$$20 \left[1 + 0.15 \left(\frac{x_1}{4400} \right)^4 \right] = 10 \left[1 + 0.15 \left(\frac{7000 - x_1}{2200} \right)^4 \right]$$

- $x_1 = 3376$ veh, $x_1 = 3624$ veh
- $t_1 = t_2 = 21$ minutes

Note: Fig 4.4 in the BLU book has an error (route 1 and route 2 graphs are swapped)

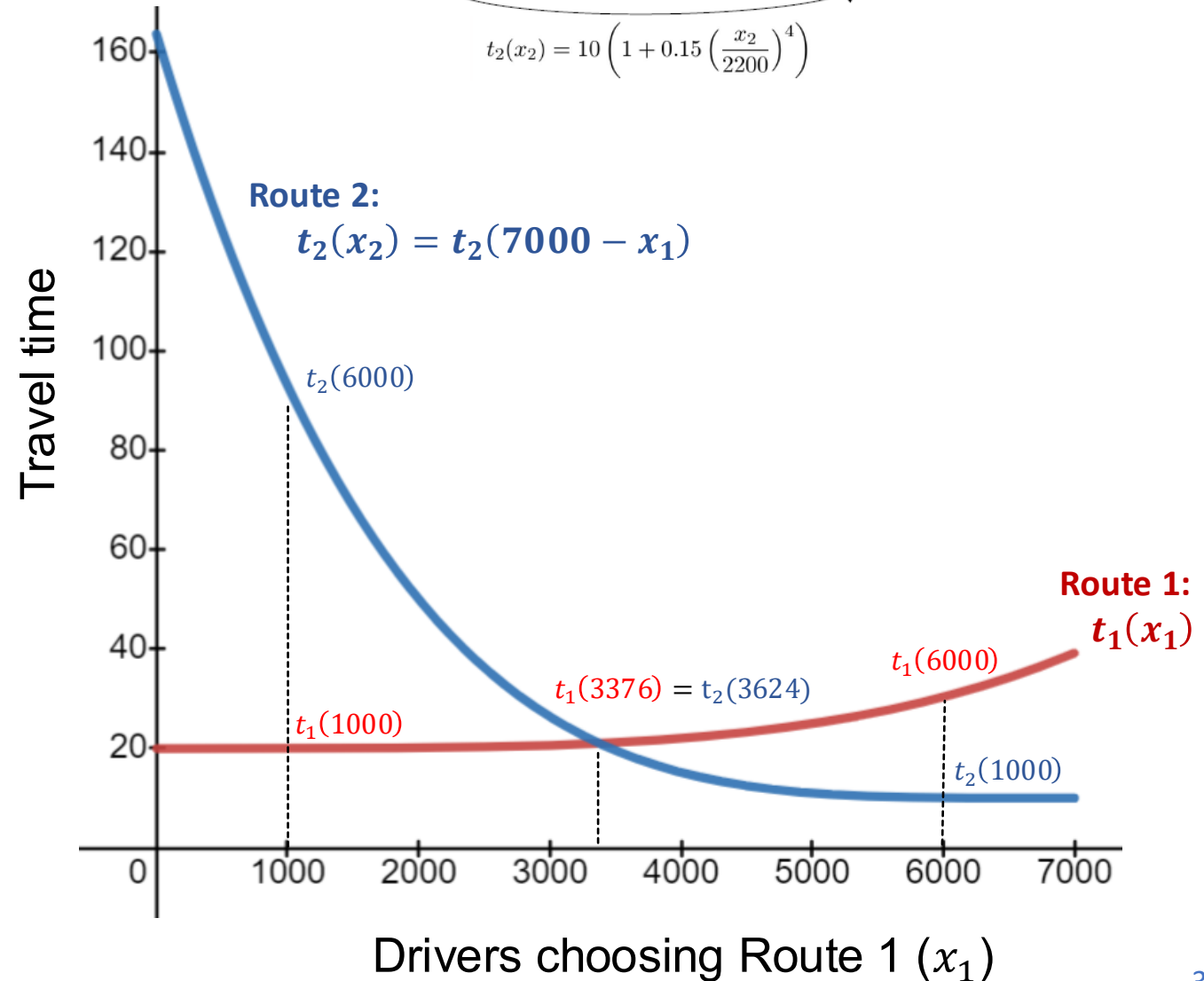
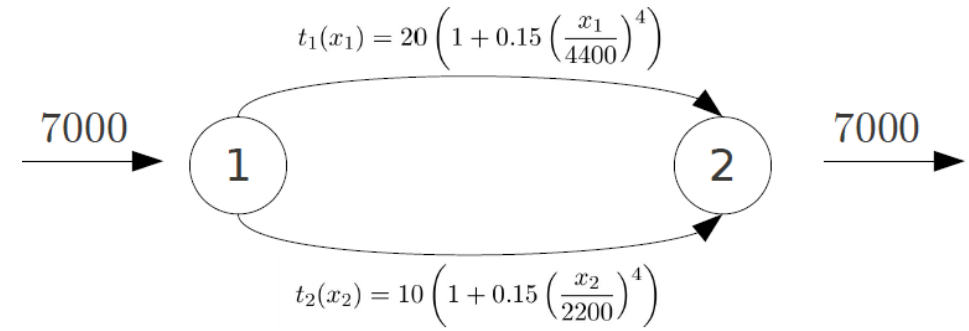
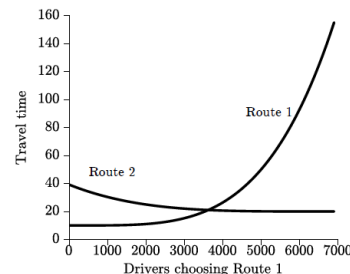
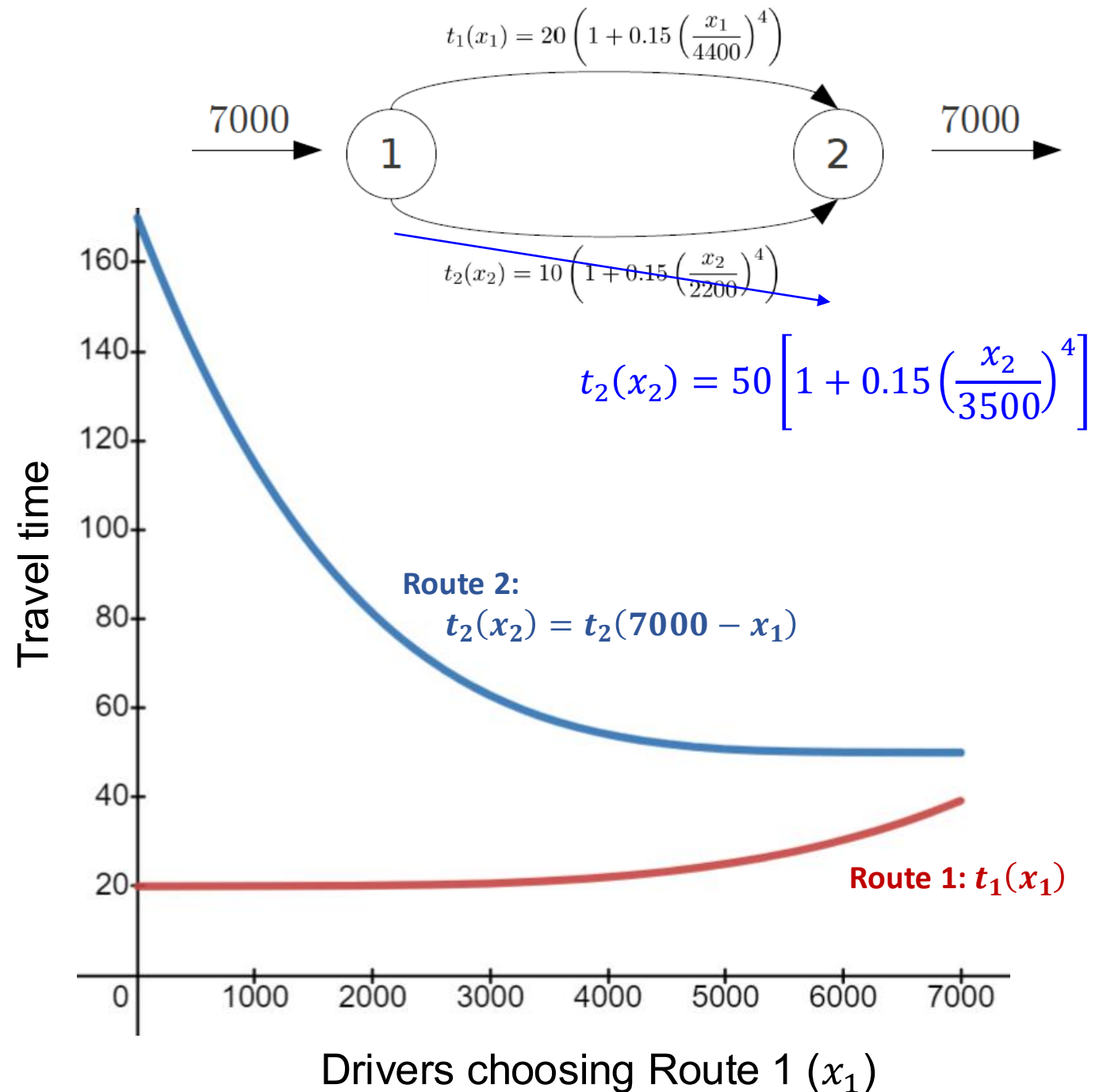


Figure 4.4: The equilibrium point lies at the intersection of the delay functions.

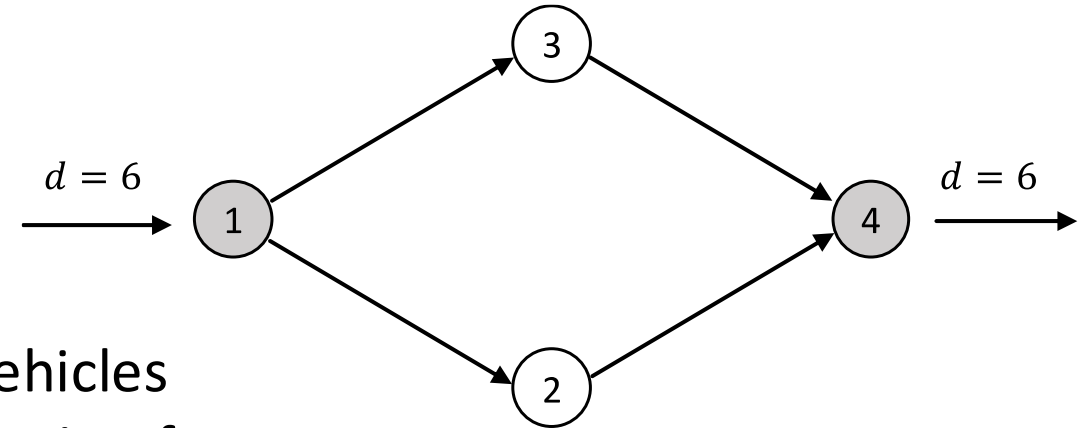
Example 4 : Modify $t_2(x_2)$

- ▶ Solving $t_1(x_1) = t_2(7000 - x_1)$ leads to *complex* solutions with imaginary parts, not *real* numbers.
- ▶ Physical interpretation
 - 'There is no way to assign 7000 vehicles to these two paths so they have equal travel times.'
- ▶ The correct UE solution is:
 - $x_1 = 7000$ veh, $x_2 = 0$ veh
 - $t_1 = 39$ minutes, $t_2 = 50$ minutes



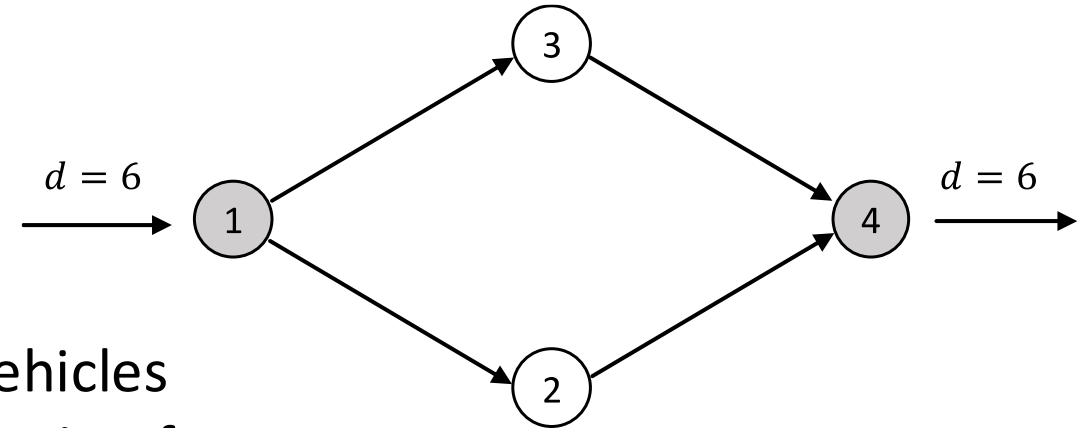
Exercise:

- Consider the network shown on the right. If 6 vehicles travel from node 1 to node 4, what is the UE solution for traffic assignment?



Link performance function	Path definition
$\begin{aligned}t_{12}(x_{12}) &= 10x_{12} \\ t_{24}(x_{24}) &= 50 + x_{24} \\ t_{13}(x_{13}) &= 50 + x_{13} \\ t_{34}(x_{34}) &= 10x_{34}\end{aligned}$	$\begin{aligned}\text{Path 1: } \pi_1 &= [1, 2, 4] \\ \text{Path 2: } \pi_2 &= [1, 3, 4]\end{aligned}$

Exercise:



- Consider the network shown on the right. If 6 vehicles travel from node 1 to node 4, what is the UE solution for traffic assignment?

- Demand constraint: $x_{13} + x_{12} = x_{34} + x_{24} = 6$
- Flow conservation: $x_{13} = x_{34}, x_{12} = x_{24}$
- UE condition: $t_{13} + t_{34} = t_{12} + t_{24}$

$$(50 + x_{13}) + 10x_{34} = 10x_{12} + (50 + x_{24})$$

$$(50 + x_{13}) + 10x_{13} = 10x_{12} + (50 + x_{12})$$

$$(50 + x_{13}) + 10x_{13} = 10(6 - x_{13}) + (50 + 6 - x_{13})$$

$$22x_{13} = 66 \rightarrow x_{13} = 3, x_{34} = 3, x_{12} = 3, x_{24} = 3$$

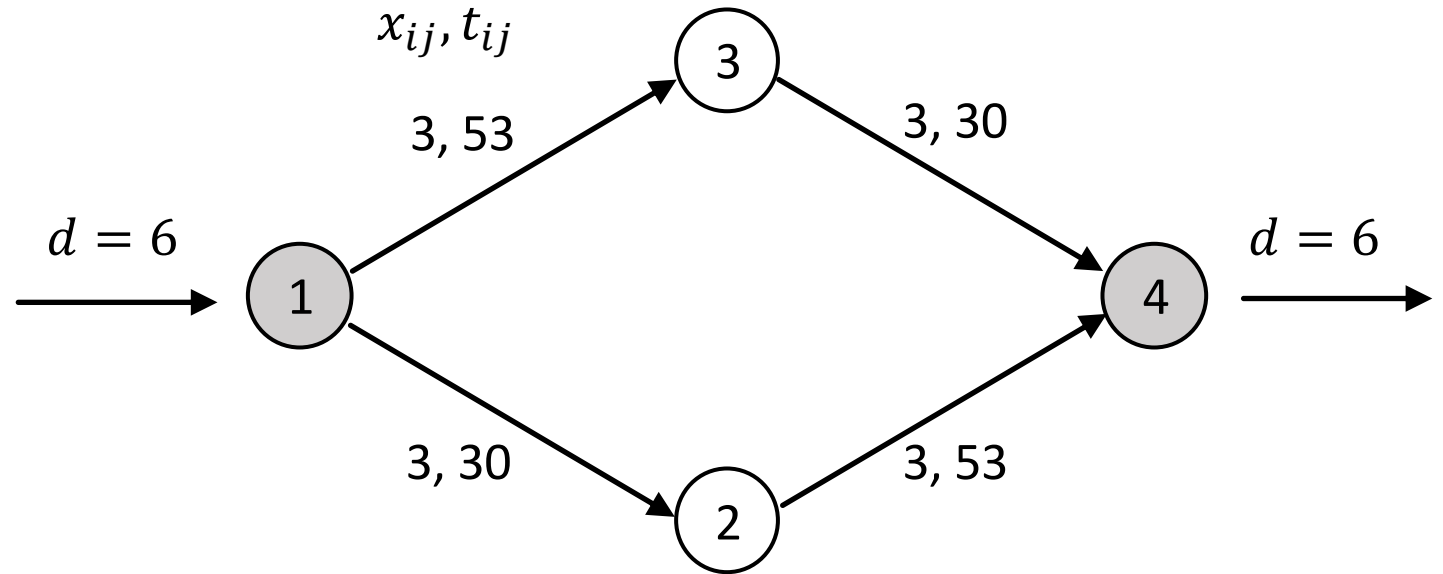
$$t_{13} = 53, t_{34} = 30, t_{12} = 30, t_{24} = 53$$
- Path travel times: $c_1 = t_{12} + t_{24} = 83, c_2 = t_{13} + t_{34} = 83$

Link performance function	Path definition
$t_{12}(x_{12}) = 10x_{12}$ $t_{24}(x_{24}) = 50 + x_{24}$ $t_{13}(x_{13}) = 50 + x_{13}$ $t_{34}(x_{34}) = 10x_{34}$	Path 1: $\pi_1 = [1, 2, 4]$ Path 2: $\pi_2 = [1, 3, 4]$

Braess's Paradox

► The UE Solution

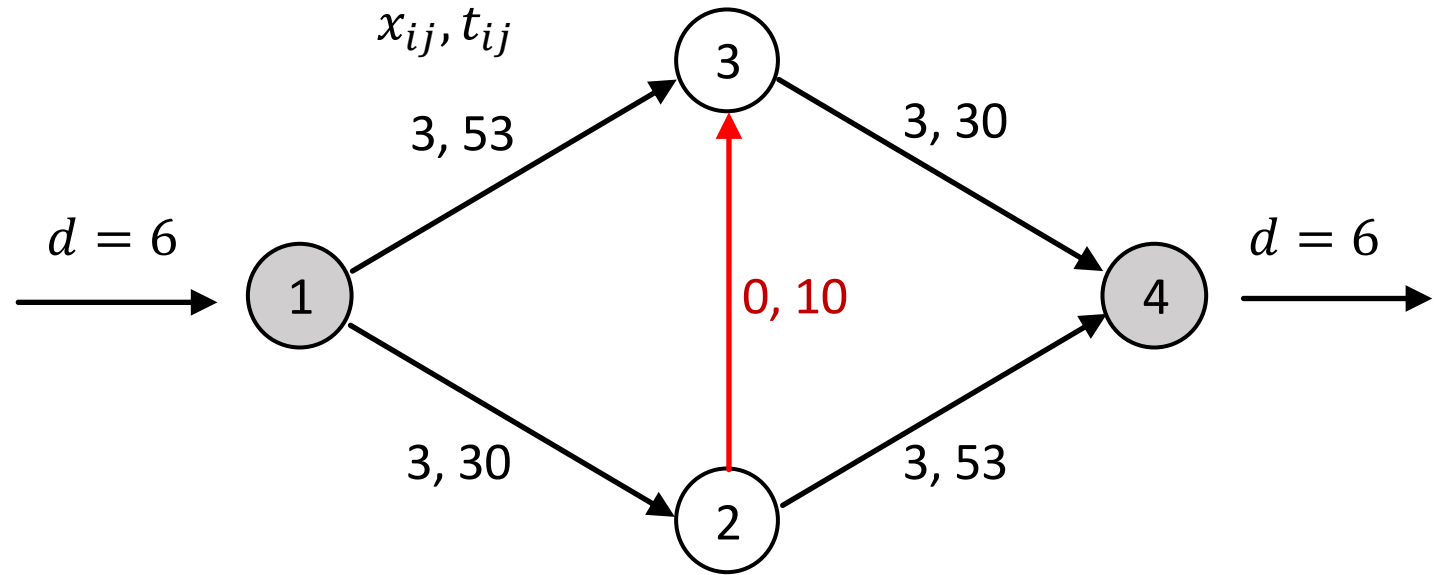
- $h_1 = 3, h_2 = 3$
- $c_1 = t_{12} + t_{24} = 30 + 53 = 83$
- $c_2 = t_{13} + t_{34} = 53 + 30 = 83$



Link performance function	Path definition
$t_{12}(x_{12}) = 10x_{12}$ $t_{24}(x_{24}) = 50 + x_{24}$ $t_{13}(x_{13}) = 50 + x_{13}$ $t_{34}(x_{34}) = 10x_{34}$	Path 1: $\pi_1 = [1, 2, 4]$ Path 2: $\pi_2 = [1, 3, 4]$

Braess's Paradox

- Assume now that the local transportation authority decides to expand the network to improve flow and reduce delay by building a new highway connecting the two intermediate nodes.



- Network performance:

- $h_1 = 3, h_2 = 3, h_3 = 0$
- $c_1 = 83, c_2 = 83, c_3 = 70$

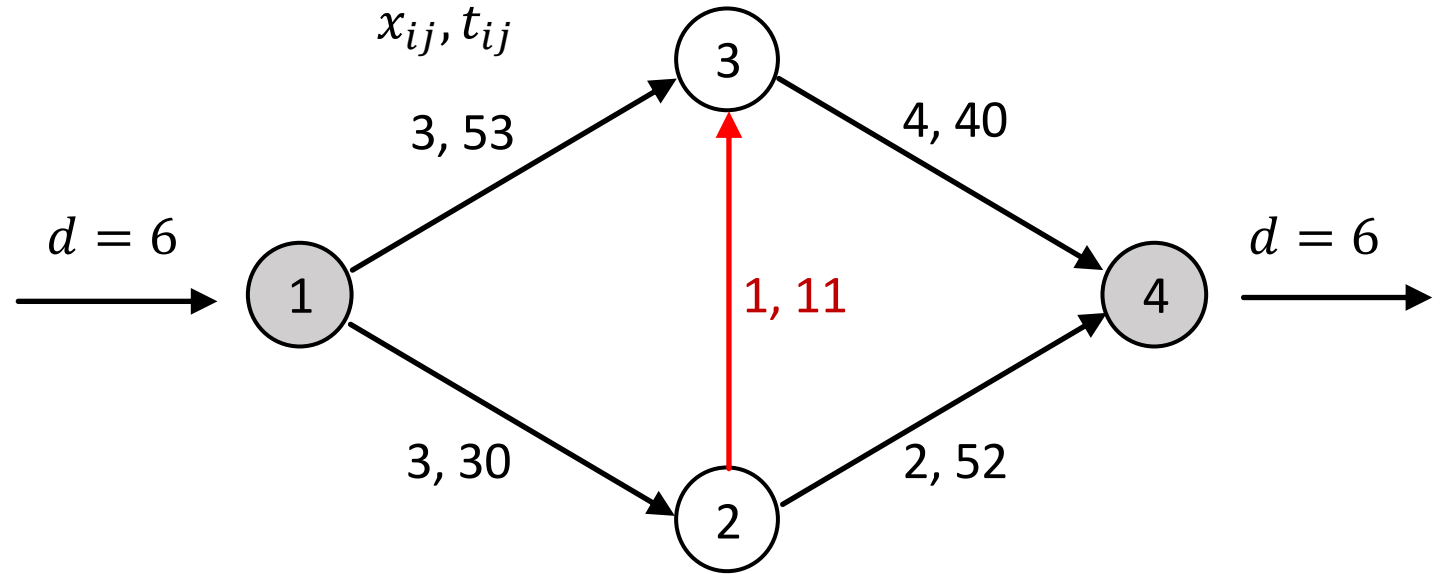
Link performance function	Path definition
$t_{12}(x_{12}) = 10x_{12}$ $t_{24}(x_{24}) = 50 + x_{24}$ $t_{13}(x_{13}) = 50 + x_{13}$ $t_{34}(x_{34}) = 10x_{34}$ $t_{23}(x_{23}) = 10 + x_{23}$	Path 1: $\pi_1 = [1, 2, 4]$ Path 2: $\pi_2 = [1, 3, 4]$ Path 3: $\pi_3 = [1, 2, 3, 4]$

Braess's Paradox

► One vehicle switch from Path 1 to Path 3

► $h_1 = 2, h_2 = 3, h_3 = 1$

► $c_1 = 82, c_2 = 93, c_3 = 81$



Link performance function	Path definition
$t_{12}(x_{12}) = 10x_{12}$ $t_{24}(x_{24}) = 50 + x_{24}$ $t_{13}(x_{13}) = 50 + x_{13}$ $t_{34}(x_{34}) = 10x_{34}$ $t_{23}(x_{23}) = 10 + x_{23}$	Path 1: $\pi_1 = [1, 2, 4]$ Path 2: $\pi_2 = [1, 3, 4]$ Path 3: $\pi_3 = [1, 2, 3, 4]$

Braess's Paradox

- ▶ One vehicle switch from Path 1 to Path 3

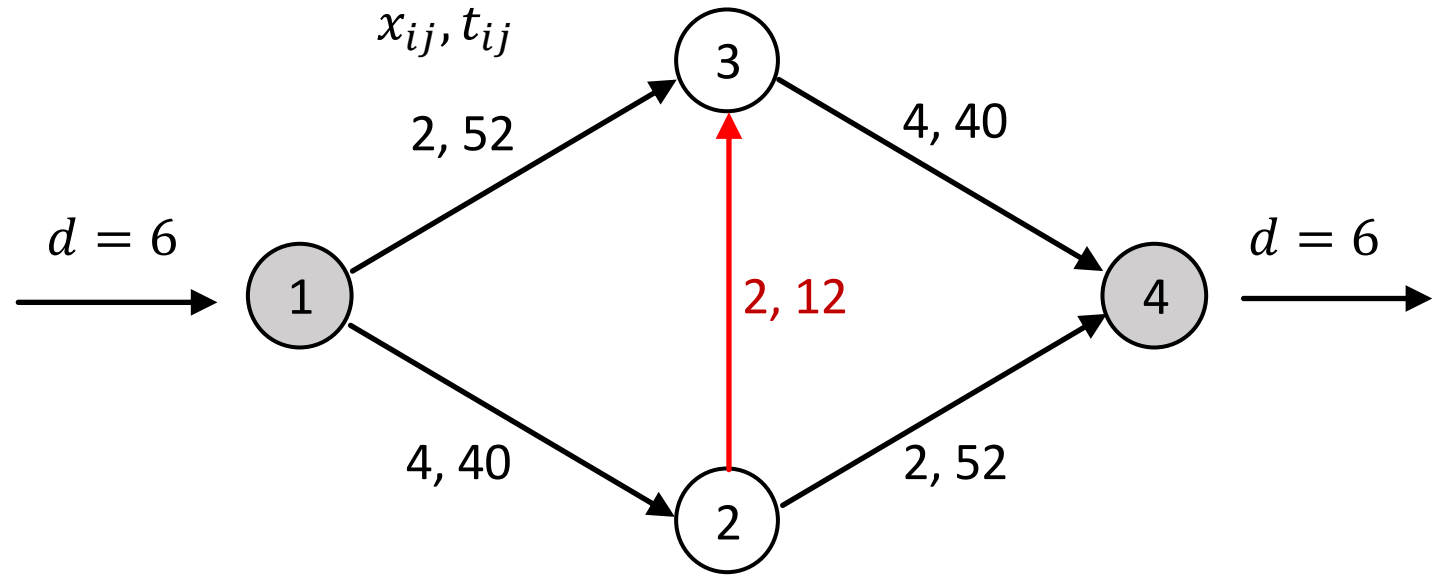
- $h_1 = 2, h_2 = 3, h_3 = 1$

- $c_1 = 82, c_2 = 93, c_3 = 81$

- ▶ One vehicle switch from Path 2 to Path 3

- $h_1 = 2, h_2 = 2, h_3 = 2$

- $c_1 = 92, c_2 = 92, c_3 = 92$



Link performance function	Path definition
$t_{12}(x_{12}) = 10x_{12}$ $t_{24}(x_{24}) = 50 + x_{24}$ $t_{13}(x_{13}) = 50 + x_{13}$ $t_{34}(x_{34}) = 10x_{34}$ $t_{23}(x_{23}) = 10 + x_{23}$	Path 1: $\pi_1 = [1, 2, 4]$ Path 2: $\pi_2 = [1, 3, 4]$ Path 3: $\pi_3 = [1, 2, 3, 4]$

Braess's Paradox

- ▶ One vehicle switch from Path 1 to Path 3

- $h_1 = 2, h_2 = 3, h_3 = 1$

- $c_1 = 82, c_2 = 93, c_3 = 81$

- ▶ One vehicle switch from Path 2 to Path 3

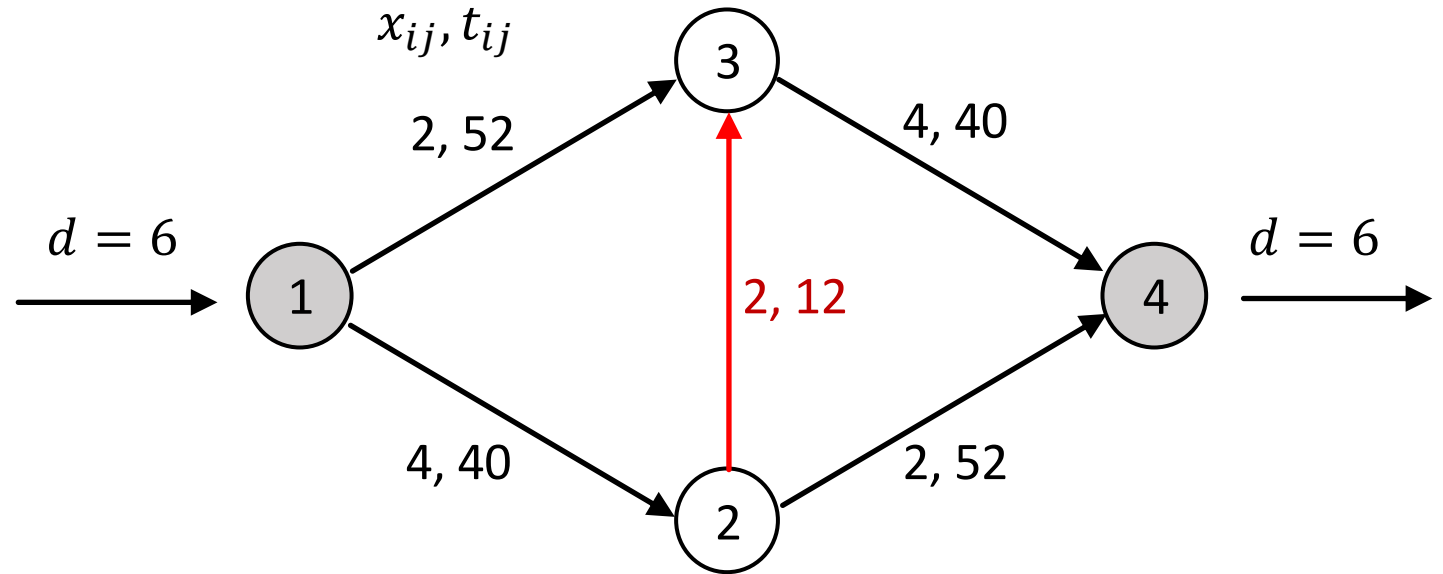
- $h_1 = 2, h_2 = 2, h_3 = 2$

- $c_1 = 92, c_2 = 92, c_3 = 92$

- ▶ The UE solution is found, but everyone is worse off!!

- Before: $c = 83$

- After: $c = 92$



Link performance function	Path definition
$t_{12}(x_{12}) = 10x_{12}$ $t_{24}(x_{24}) = 50 + x_{24}$ $t_{13}(x_{13}) = 50 + x_{13}$ $t_{34}(x_{34}) = 10x_{34}$ $t_{23}(x_{23}) = 10 + x_{23}$	Path 1: $\pi_1 = [1, 2, 4]$ Path 2: $\pi_2 = [1, 3, 4]$ Path 3: $\pi_3 = [1, 2, 3, 4]$

Braess's Paradox: Implications

- ▶ Adding extra capacity to a network does not always reduce traffic congestion.
 - Building a new roadway link can actually worsen travel times for all travellers in the network, even though the demand is the same!
- ▶ User equilibrium does *not* minimise congestion.
 - Each driver minimises his or her *own* travel time.
 - The individual choice of route is carried out with no consideration of the effect of this action on other network users.

System Optimisation (SO)

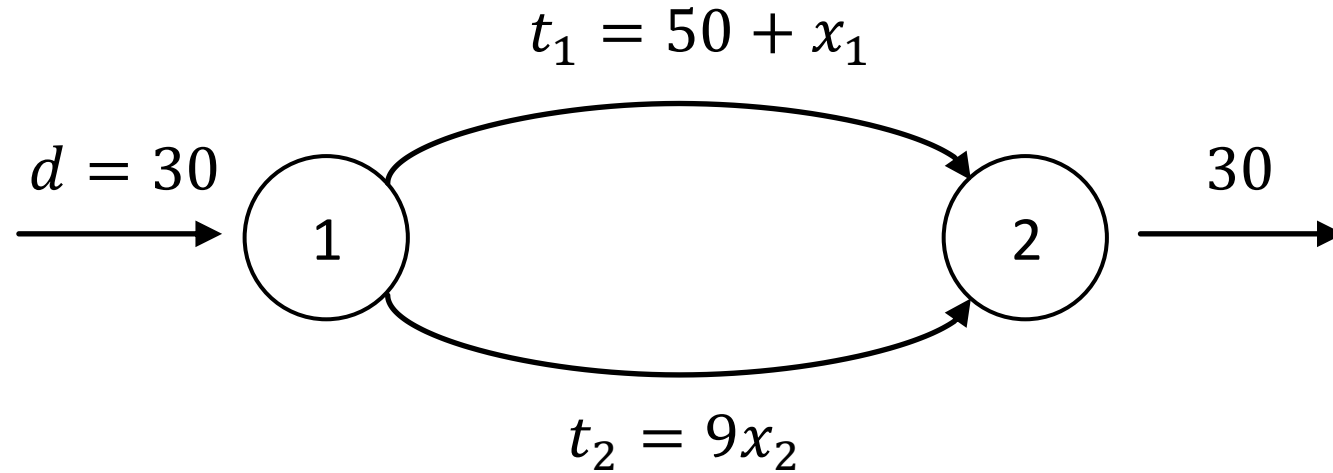
Route Assignment Rule

System Optimal (SO) Assignment Rule

- ▶ The **System Optimal** (SO) assignment is another possible assignment rule that results in the *system optimal* state where the *total system travel time* (TSTT) is minimised.
- ▶ The **total system travel time** is defined as:

$$TSTT = \sum_{\pi \in \Pi} h^{\pi} c^{\pi} = \sum_{(i,j) \in A} x_{ij} t_{ij}$$

Example 5



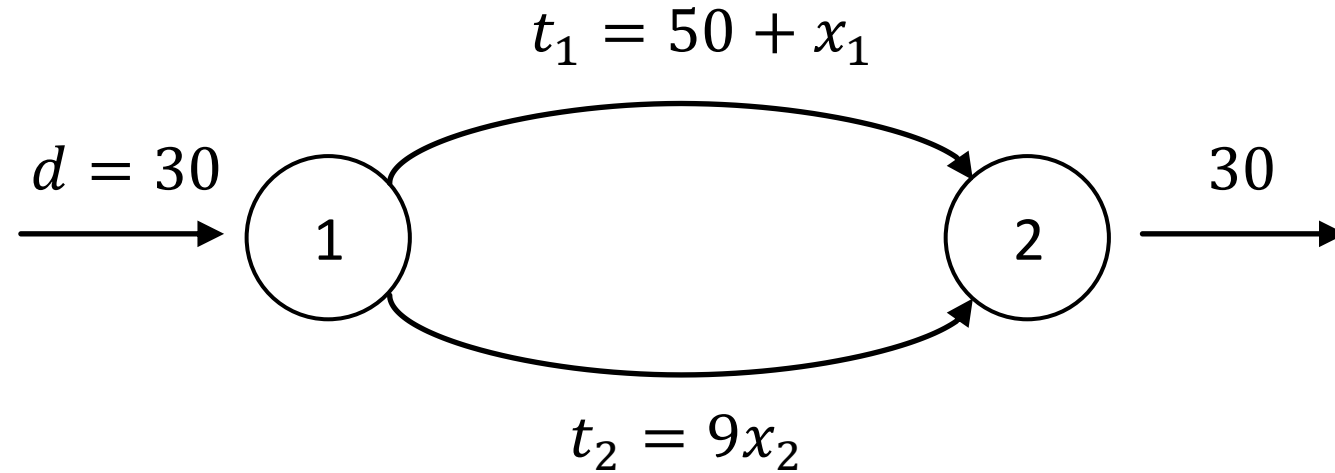
- Write the expression for TSTT :

- $$TSTT = \sum_{(i,j) \in A} x_{ij} t_{ij} = x_1(50 + x_1) + x_2(9x_2) = 50x_1 + x_1^2 + 9x_2^2$$
$$= 50x_1 + x_1^2 + 9(30 - x_1)^2 = 10x_1^2 - 490x_1 + 8100$$

- This is a quadratic function, whose minimum can be found by solving $\frac{dTSTT}{dx_1} = 0$:

- $$\frac{dTSTT}{dx_1} = 20x_1 - 490 = 0 \rightarrow x_1 = 24.5, x_2 = 5.5 \rightarrow t_1 = 74.5, t_2 = 49.5$$

Example 5



UE solution (from Example 2)	SO solution
$x_1 = 22, x_2 = 8$ $t_1 = 72, t_2 = 72$ $TSTT = 22(72) + 8(72) = 2160$	$x_1 = 24.5, x_2 = 5.5$ $t_1 = 74.5, t_2 = 49.5$ $TSTT = 24.5(74.5) + 5.5(49.5) = 2097.5$

Notes on System Optimal (SO) Assignment

- ▶ The SO solution often does not coincide with the UE solution.
- ▶ The SO state is generally not an equilibrium flow pattern, since some travellers can be better off by unilaterally changing routes.
- ▶ The SO flow pattern does not reflect actual, realistic travel behaviour.
- ▶ **The value of the SO formulation:**
 - It provides a **theoretical lower bound** on the delay in a network
 - It serves as a **yardstick/benchmark for the best performance** by which other more realistic assignment rules can be compared to

Static Traffic Assignment (STA) and Dynamic Traffic Assignment (DTA)

Traffic Assignment Problem

- ▶ Traffic assignment models are largely categorised into **Static Traffic Assignment** (STA) models and **Dynamic Traffic Assignment** (DTA) models.
- ▶ The difference between static and dynamic traffic assignment lies in the **traffic flow models** used.

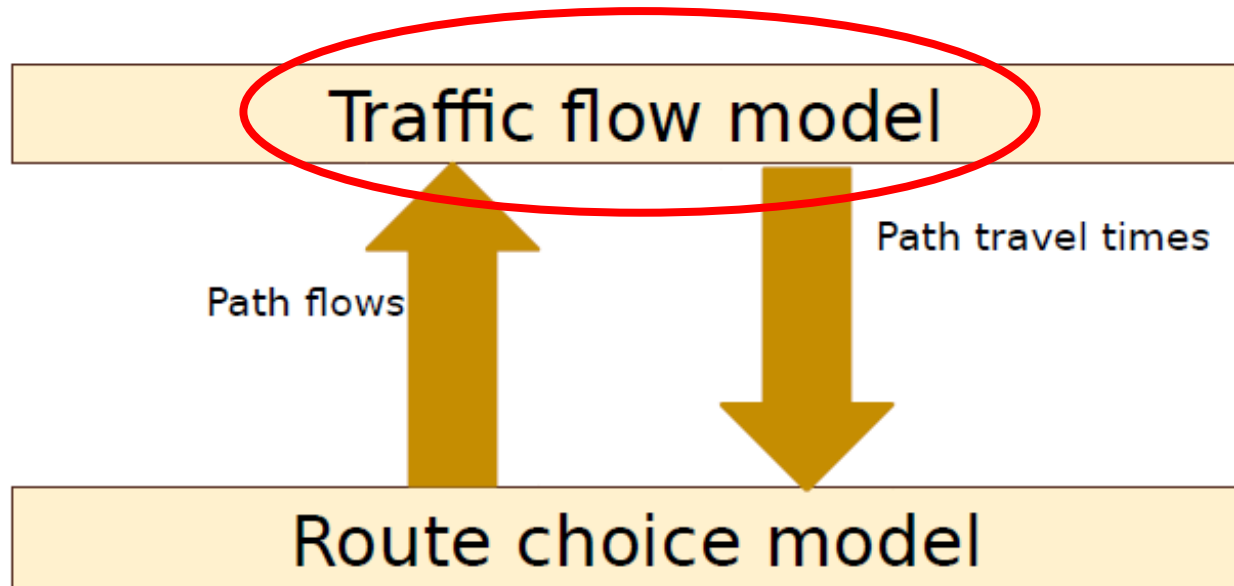
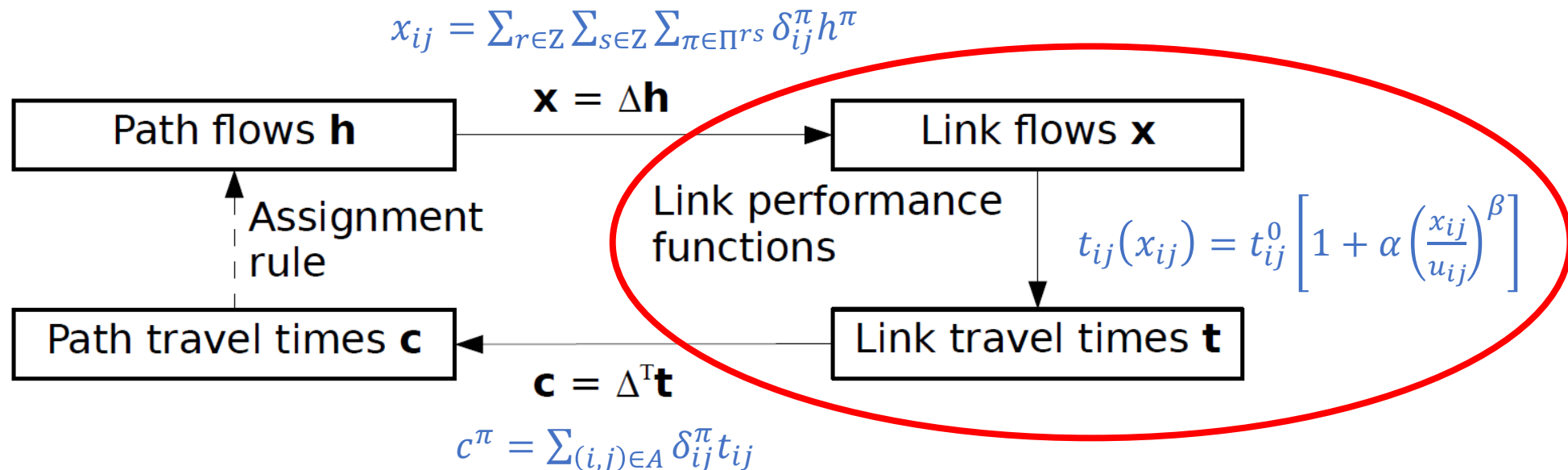


Figure 1.3: Traffic assignment as interacting systems.

Static Traffic Assignment (STA)

- ▶ In STA, the traffic flow model is based on **analytical link performance functions** (e.g., BPR function) to map traffic flow (demand) to congestion.
- ▶ Assumes a **steady-state** condition where traffic **demand** and **congestion** remain constant over time.



Static Traffic Assignment (STA)

► Benefits of STA

➤ Mathematical tractability:

- The STA problem can be formulated using well-established mathematical tools of optimisation
- The existence and uniqueness of UE solutions can be assured under mild conditions.
- The results produced by STA models are intuitive and easy to interpret.

➤ Computational efficiency:

- STA is relatively easy to implement, requiring fewer data and computational resources.
- STA can produce results quickly, even in realistic networks involving tens of thousands of links

Static Traffic Assignment (STA)

► Limitations of STA

➤ Static nature:

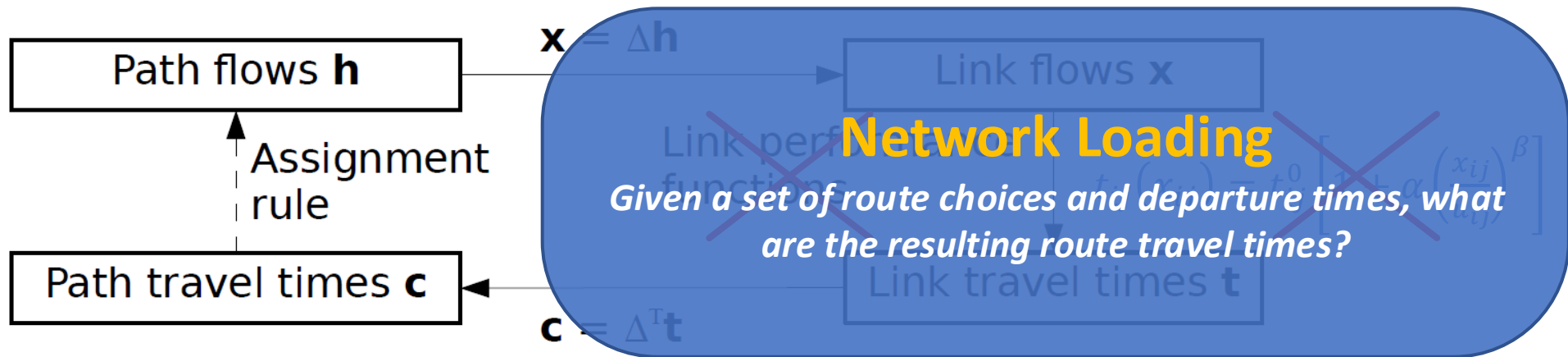
- STA assumes that traffic demand and network congestion remain constant over time, which is often not the case in real-world transport networks.
 - Travelers choose different departure times, i.e., traffic demand changes across the time of the day.
 - Travelers departing at different times may experience different travel times

➤ Inaccuracy:

- Common link performance functions like the BPR function are highly simplified and less realistic.
- STA does not capture *queue spillback* phenomena (traffic to fill the entire length of a link, causing the queue to spill over onto upstream links)

Dynamic Traffic Assignment (DTA)

- ▶ DTA uses more realistic traffic flow models that can track changes in congestion and link travel times over time (at a fairly fine resolution, on the order of a few seconds).
- ▶ The mapping from path/link flows to path/link travel times is no longer a simple link performance function but a complex process called '**network loading**'.

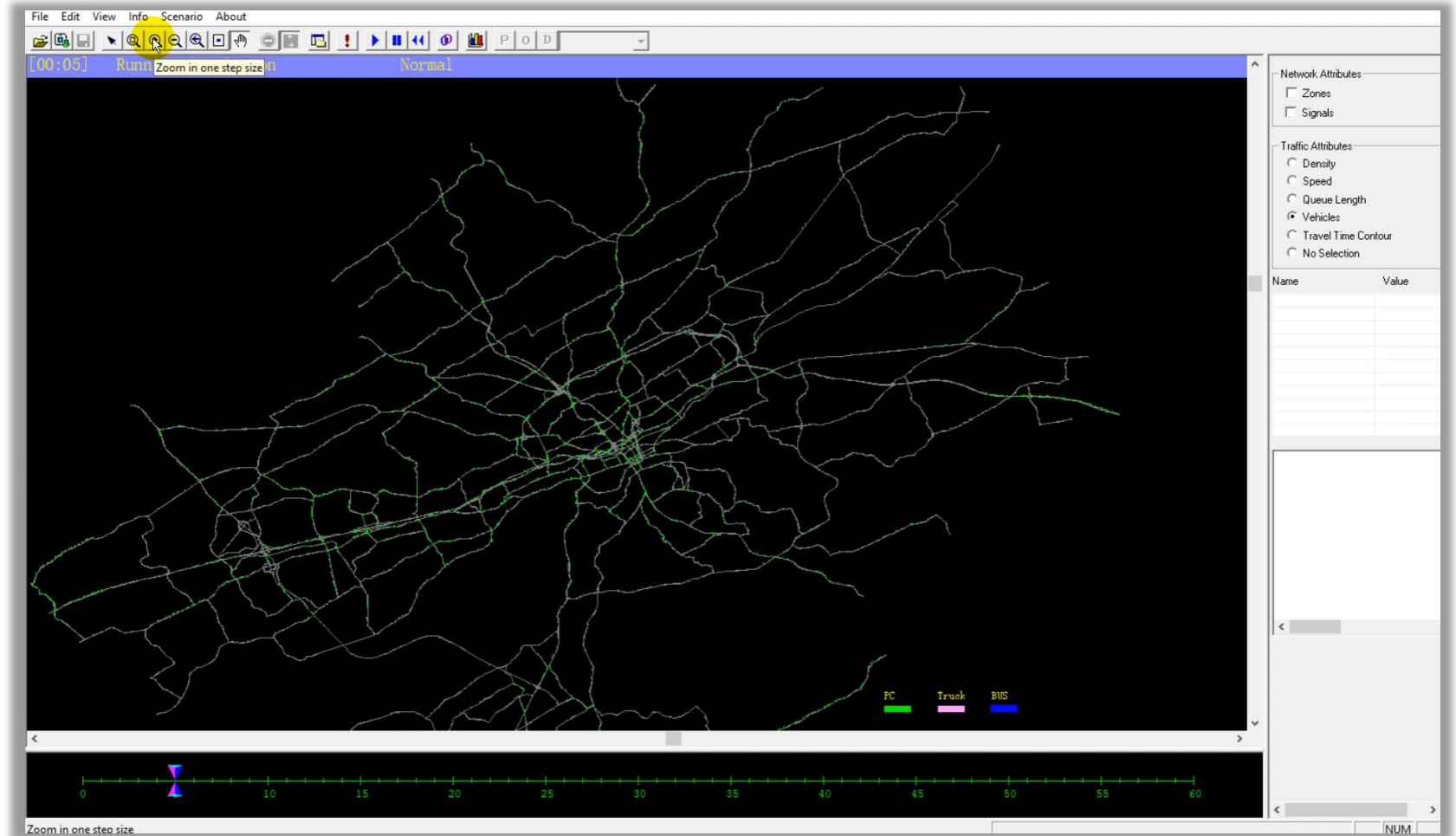


Network Loading Approaches

- ▶ **Network Loading:** predicts how traffic propagates in the network, given the routes and departure times of all drivers as inputs
- ▶ ***Simulation-based network loading:*** uses **computer simulation (traffic simulation models)** to replicate the network-wide movements of a large number of vehicles over time and capture their spatial and temporal interactions
 - **Simulation-based DTA models**
 - **Traffic simulation models**
 - Macroscopic
 - Mesoscopic
 - Microscopic

Simulation-based DTA Model (Mesoscopic)

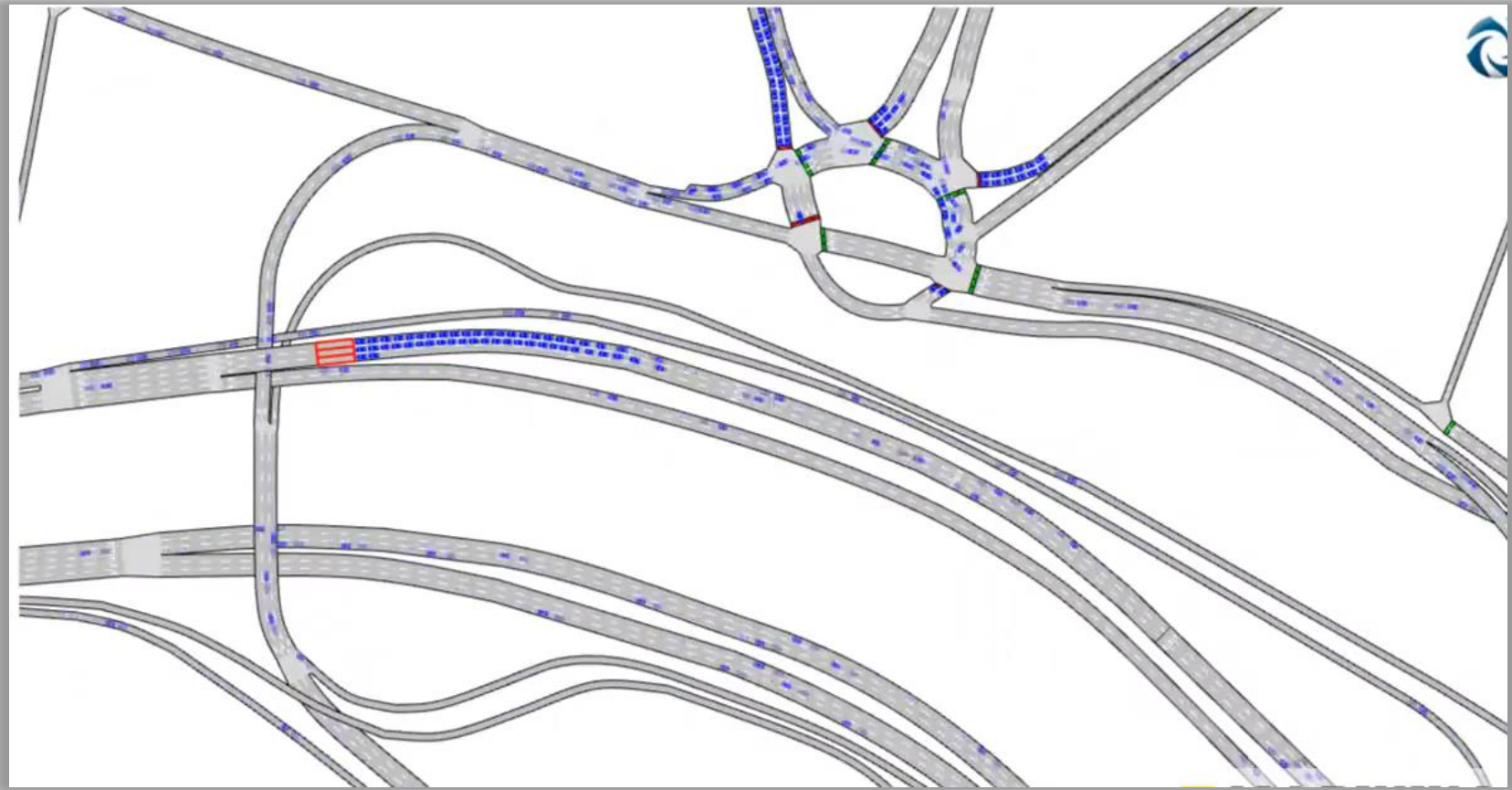
- ▶ Mesoscopic traffic simulation-assignment model
- ▶ Model individual vehicles based on **link-level, aggregate traffic flow models (i.e., fundamental diagrams)**
 - The relation between link density (number of vehicles) and link speed (congestion level)



DYNASMART Mesoscopic Traffic Simulation-Assignment Model

Simulation-based DTA Models (Microscopic)

- ▶ Microscopic traffic simulation model
- ▶ Model individual vehicles based on **disaggregate, microscopic traffic flow models (i.e., driving behaviour models)**
 - car-following model, lane-changing model



Aimsun Microscopic Traffic Simulation-Assignment Model

Thank you